Selective Admissions and Academic Value-Added

Evidence from Indian Higher Education

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Abstract

We estimate the academic value-added of college–major programs using centralized admissions data from a large higher-education system in India. The setting combines a standardized entrance examination, preference-based allocation, and common exit examinations, allowing us to measure learning on a common scale and identify effects at program-specific admission cutoffs using a regression discontinuity (RD) design. Admission to more selective programs improves academic achievement by 0.11 standard deviations and raises graduation rates by 3.5 percentage points for marginal admits. We compare these RD estimates to observational, ordinary least squares (OLS) value-added measures constructed with rich baseline controls. While OLS and RD estimates align at central admission margins, they diverge sharply at affirmative-action cutoffs. Leveraging multiple cutoffs within the same program, we develop a framework that decomposes value-added into an academic-environment component and an ordinal-rank component. Entering more selective programs lowers relative rank, generates improvements in students' peer environment, and alters the exposure to institutional quality. Our findings show that in selective admissions systems, academic value-added is inherently margin-specific, reflecting the joint effects of changes in peer composition, institutional exposure, and ordinal rank induced by admission.

Keywords: College Quality, Academic Value-Added, Regression Discontinuity, Ordinal Rank Effects, Affirmative Action

JEL Codes: I21, I23, I24, C31

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1 Introduction

Selective colleges and fields of study have large effects on students' later labor-market outcomes (Hoekstra, 2009; Zimmerman, 2014; Kirkeboen et al., 2016), and a substantial body of research has therefore used these outcomes to study the value of attending particular programs.[1] Labor-market measures, however, reflect a combination of channels—including signaling, networks, and bargaining—and therefore may not isolate the academic contribution that programs make to student learning. In addition, much of the higher-education literature has concentrated on highly selective institutions (Dale and Krueger, 2002, 2014; Black and Smith, 2004, 2006), where sharp selection margins and rich data are most readily available, rather than on the broader system in which most students are enrolled. Concurrently, selection into colleges and affirmative-action policies shape heterogeneity in the margins at which different social groups enter particular programs (Bertrand et al., 2010; Bagde et al., 2016; Arcidiacono and Lovenheim, 2016). Taken together, these considerations motivate the need for credible measures of program-level academic quality across a wide network of college-major programs, beyond the most selective segments of higher education.

Measuring program-level academic quality, however, is challenging. Students sort into majors based on prior preparation, preferences, and institutional constraints, making it difficult to disentangle program effects from selection (Card, 1999; Altonji et al., 2012; Kirkeboen et al., 2016). Observational value-added models attempt to adjust for this sorting but may remain biased, particularly for disadvantaged students whose preparation and constraints differ systematically (Rothstein, 2009; Cunha and Miller, 2014; Mountjoy and Hickman, 2021). A further complication is the lack of comparability in academic measures across programs and institutions.[2] More fundamentally, few higher-education systems combine the institutional features required to credibly measure academic value-added at scale: a standardized entrance exam, a centralized preference-based admissions mechanism, and standardized exit examinations administered uniformly across colleges. We leverage a setting that uniquely combines these features, creating a rare opportunity to measure program-level academic value-added across an entire higher-education network.

In this paper, we estimate program-level academic value-added using two distinct approaches. First, we construct observational value-added measures for each program using Ordinary Least Squares (OLS) specifications with rich controls. Second, we estimate effects at admission cutoffs using a regression discontinuity (RD) design.

[1] A related literature examines academic outcomes directly—often in selective school or university contexts (Abdulkadiroğlu et al., 2014)—but this evidence remains limited relative to the extensive literature on labor-market returns.

[2] In most higher-education systems, admissions scores and academic outcomes are measured on program-specific scales rather than through common entry and exit examinations, limiting cross-program comparability.

The RD framework identifies the local average treatment effect (LATE) of marginal admission into a program, while OLS summarizes average conditional differences across enrolled students within programs. Comparing these estimators within the same institutional setting allows us to characterize how admission margins shape commonly used value-added measures and why OLS and RD summarize different counterfactual comparisons.

A central reason OLS and RD estimates may diverge is that cutoff-based comparisons move students into more selective cohorts where they occupy different ordinal positions within the destination program. Admission at the margin can therefore generate ordinal-rank effects that offset or amplify the effect of academic environment (Fabregas, 2023; Elsner and Isphording, 2017; Murphy and Weinhardt, 2020; Denning et al., 2023). Leveraging multiple within-program cutoffs generated by affirmative action policies, we develop a framework that separately identifies ordinal-rank effects from other components of the academic environment.

Our empirical setting is a centralized admissions system governing entry into undergraduate engineering college–major programs. A single standardized entrance exam and a preference-based allocation algorithm determine placement across colleges and majors, with distinct cutoffs for each affirmative action group and for General-category students (Baswana et al., 2019). These program-by-category thresholds generate multiple sharp regression discontinuity designs across institutions and fields of study, enabling credible identification of causal effects at heterogeneous admission margins. We link admissions records to standardized academic outcomes, including semester examinations administered uniformly across colleges, allowing consistent measurement of academic learning on a common scale. Our empirical strategy combines observational value-added models (Chetty et al., 2014; Cunha and Miller, 2014; Koedel and Rockoff, 2015; Mountjoy and Hickman, 2021) with RD-based causal estimates inspired by the selective-school and cutoff-based admissions literatures (Hoekstra, 2009; Pop-Eleches and Urquiola, 2013; Abdulkadiroğlu et al., 2014; Zimmerman, 2014), drawing on standard regression-discontinuity methods (Hahn et al., 2001; Imbens and Lemieux, 2008; Lee and Lemieux, 2010).

We estimate program-level causal effects by exploiting sharp program-by-category admission cutoffs that generate locally random variation in program entry at the margin. Crossing a cutoff produces a large and discrete increase in the probability of enrollment—from near zero to roughly 0.88—indicating a strong first stage and supporting a LATE interpretation. Within each bandwidth, we compare marginally admitted students to marginally rejected students who submitted identical preference lists, ensuring that differences in outcomes are not driven by differences in program targeting or stated choice sets. Using this design, we find that admission to more selective programs increases standardized examination scores by approximately 0.11 standard

deviations and raises graduation rates by about 3.5 percentage points. These gains are not confined to a single major.[3]

To benchmark these causal estimates against conventional practice, we also construct observational value-added (VA) measures for all programs using rich student-level controls. Comparing OLS and RD estimates reveals systematic differences that depend on the admission margin being evaluated. For General-category students—whose admission thresholds are typically closest to the program median—observational value-added aligns closely with RD-based estimates, with similar median effects across programs. In contrast, for students admitted through affirmative-action cutoffs, who enter programs at substantially lower ordinal ranks within the destination cohort, OLS and RD estimates diverge sharply, with OLS often indicating negative value-added where RD estimates remain positive.[4]

These patterns indicate that OLS and RD are not simply more or less biased estimators of a common parameter; rather, they correspond to different counterfactual comparisons because students enter programs at distinct admission margins. Interpreting program-level value-added therefore requires explicit attention to the margin underlying each estimate.

To unpack the mechanisms underlying these differences, we decompose the academic environment into two components: (i) a program-level academic environment that bundles peer quality and institutional inputs, and (ii) a student's ordinal rank within the cohort. Admission to a more selective program typically improves peer quality and institutional resources, channels expected to raise performance. At the same time, entering a stronger cohort lowers a student's relative standing compared to their next-best alternative.

Empirically, marginal admittees experience a 0.39 standard deviation improvement in peer quality but an 18 percentile-point decline in ordinal rank within the destination cohort. Exploiting variation across category-specific cutoffs within programs, we estimate that this decline in ordinal rank reduces academic performance by approximately 0.24 standard deviations. In contrast, the residual academic-environment component—capturing improvements in peer composition and institutional inputs—is strongly positive, with a median effect of roughly 0.44 standard deviations. The positive net RD value-added therefore reflects that improvements in the academic environment more than offset the penalty from lower relative rank.

Our work contributes to three strands of the literature. First, it advances research

[3]Because semester examinations are standardized within, but not across, engineering majors, our primary specifications are estimated separately by major. We find similar RD patterns across the largest engineering majors, suggesting that the positive effects are not driven solely by one major such as Computer Science.

[4]These divergences are most pronounced in highly selective programs, where admission margins differ most sharply and ordinal-rank shifts are largest.

on college quality and the measurement of program effectiveness (Black and Smith, 2004, 2006; Hoekstra, 2009; Zimmerman, 2014; Cunha and Miller, 2014; Sekhri, 2020; Mountjoy and Hickman, 2021). Existing studies evaluate institutions using labor-market outcomes and often focus on selective settings where identification is sharp but coverage is narrow. We instead leverage a system that combines standardized entry exams, centralized admissions, and common exit examinations to estimate academic learning value-added across an entire higher-education network. Moreover, by comparing observational (OLS) and regression-discontinuity (RD) value-added within the same institutional setting, we clarify how heterogeneous admission margins generate systematic differences between these measures and how such differences reshape program rankings.

Second, we contribute to the literature on ordinal rank effects (Murphy and Weinhardt, 2020; Elsner and Isphording, 2017; Elsner et al., 2021; Delaney and Devereux, 2022; Dasgupta et al., 2022; Denning et al., 2023; Fabregas, 2023). Prior work documents how a student's relative position within a cohort shapes academic performance, major choice, preferences, and longer-run outcomes. However, this literature has not been directly integrated with causal measures of program quality in large, multi-institution admissions systems. Exploiting multiple admission cutoffs within programs, we identify ordinal-rank effects at distinct admission margins and quantify their contribution to overall program value-added.

Third, we engage the affirmative action and mismatch literatures (Bagde et al., 2016; Bertrand et al., 2010; Arcidiacono et al., 2011; Arcidiacono and Lovenheim, 2016; Otero et al., 2021). We show that estimated program effects for disadvantaged students are highly sensitive to the estimator and admission margin under consideration. In settings where affirmative-action students enter at substantially lower ordinal ranks, observational value-added measures can diverge sharply from margin-specific RD estimates. Our findings caution against interpreting negative observational patterns as evidence of mismatch in the absence of margin-specific causal evidence.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting, admissions mechanism, and data. Section 3 outlines the empirical framework and identification strategy. Section 4 presents observational and regression-discontinuity estimates of program-level academic value-added and compares their implications across programs and student groups. Section 5 develops a decomposition of program value-added into ordinal-rank and residual academic-environment components. Section 6 concludes with implications for value-added measurement and the evaluation of equity-oriented admissions policies.

2 Background

2.1 Higher-Education System and Institutional Context

India operates one of the largest higher-education systems in the world. As of the 2021–22 academic year, the gross enrollment ratio (GER) for individuals aged 18–23 stood at approximately 28.5 percent, reflecting rapid expansion over the past two decades (AISHE, 2023). The system comprises 42,825 colleges nationwide, nearly four-fifths of which are privately managed, underscoring the central role of private provision in tertiary education. Within this landscape, Engineering and Technology constitutes one of the most prominent undergraduate streams. Nationally, more than 4.13 million students are enrolled in engineering and technology programs across all levels, with undergraduate B.E./B.Tech programs alone accounting for roughly 3.9 million students. Despite representing only about 6 percent of all colleges, engineering institutions absorb a disproportionate share of enrollment, reflecting both strong student demand and the sector's perceived labor-market relevance.

We study a large Indian state whose higher-education system closely mirrors national trends in scale, composition, and governance. According to the All India Survey on Higher Education (AISHE), the state has over 2,000 colleges enrolling more than 1.18 million undergraduate students at a time (AISHE, 2023). Engineering education plays a central role within this system. The state hosts approximately 175 engineering colleges offering undergraduate B.E./B.Tech programs, with a combined sanctioned intake exceeding 100,000 seats, and more than 80 percent of these institutions are privately managed (DSE, 2023). These colleges are affiliated with a common state university, resulting in a unified academic calendar, standardized curricula, and centrally administered semester examinations. This combination of scale, institutional uniformity, and standardized academic outcomes makes the state an especially suitable setting for studying program-level academic value added across a broad network of colleges rather than a narrow set of elite institutions.

2.2 Centralized Admissions, Preferences, and Reservation Rules

Common Entrance Examination. Admission to undergraduate engineering programs in the state is governed by a single, standardized Common Entrance Test (CET) administered annually by the state government. The CET is conducted jointly for multiple undergraduate streams, including engineering, agriculture, and pharmacy, with stream-specific ranks used for program allocation. The examination assesses applicants on core subjects drawn from the higher-secondary curriculum—typically mathematics, physics, and chemistry—ensuring that test scores are comparable across all candidates. A composite score is constructed using performance in the entrance ex-

amination together with marks obtained in the final year of secondary schooling. This composite score is then used to create a rank that serves as the sole academic criterion for program allocation.[5] The use of a common examination and unified ranking system establishes a single ordering of applicants across colleges and majors, forming the basis for centralized allocation.

Preference Submission and Program Choice. Following the announcement of entrance exam ranks, applicants submit a ranked list of preferences over college-major combinations. Preferences are expressed jointly over institution and field of study, allowing students to prioritize both program specialization and college quality. Preference lists are submitted prior to allocation and guide all subsequent assignment rounds. While applicants may revise their preference ordering across rounds, preferences are always declared before assignment outcomes are realized in that round. As a result, admission cutoffs emerge from aggregate applicant demand and program capacity constraints rather than from strategic targeting of particular majors during allocation.

Reservation Categories and Affirmative Action. The centralized admissions system incorporates constitutionally mandated reservation policies for historically disadvantaged groups. Seats within each college-major combination are allocated separately across reservation categories, including Scheduled Castes (SC), Scheduled Tribes (ST), and Other Backward Classes (OBC), alongside the General category.[6] Each category is assigned a fixed share of seats, and applicants compete only within their respective category for those seats. As a result, admission thresholds are defined separately for each reservation category within the same college and major, reflecting both category-specific demand and seat availability.

Allocation Algorithm and Multiple Cutoffs. Program allocation is carried out through a centralized deferred acceptance algorithm that operates independently within each reservation category (Baswana et al., 2019; Aiyer, 2025). Within a given category, applicants are considered in order of their composite rank and assigned to their most preferred available college-major combination, subject to capacity constraints. This process continues until all seats in that category are filled or all applicants have been considered. Because the algorithm runs independently across reservation categories, each college-major combination generates multiple admission cutoffs (one for each category) reflecting differences in competition and reservation-specific seat allocations.

[5]The composite score places 25% weight on higher-secondary examination performance and 75% weight on common entrance examination performance. Exact weights are fixed by the state authority and remain constant across applicants within a given admission cycle.

[6]In addition to caste-based reservations, the system includes quotas for women, persons with disabilities, and other designated groups; these operate within or alongside primary reservation categories according to state rules.

Acceptance, Reporting, and Final Enrollment. Applicants who are offered a seat through the centralized allocation process may accept the offer and report to the allotted college or decline and participate in subsequent rounds of allocation. Multiple rounds are conducted, allowing students to update acceptance decisions and, if desired, revise their preference ordering between rounds. Final enrollment reflects the outcome of this iterative process once all rounds are completed. Importantly, applicants located just above and just below a given college-major–category cutoff have faced the same set of program options and submitted preferences prior to assignment, differing only in their rank relative to the cutoff. As a result, final enrollment outcomes change discretely at program-specific thresholds while underlying eligibility rules and program offerings remain constant.

2.3 Academic Structure and Outcome Measurement

Standardized Curriculum and Examinations. All undergraduate engineering programs in the state follow a common academic structure prescribed by a single affiliating state university. Curricula are standardized within each engineering branch (e.g., Computer Science Engineering, Electronics and Communication Engineering) and are uniform across colleges, irrespective of ownership or location. Instruction is organized on a semester basis, with students progressing through a fixed sequence of courses over eight semesters. At the end of each semester, students take university-administered examinations that are identical across colleges within a given branch. Examination papers are centrally designed, evaluated using common grading schemes, and subject to uniform moderation procedures. This centralized academic framework ensures that semester examination outcomes are directly comparable across institutions offering the same program.

Degree Completion and Graduation Outcomes. In addition to semester examination performance, we observe degree completion outcomes for all admitted students. Undergraduate engineering programs are designed to be completed over four academic years, with progression governed by uniform credit and examination requirements set by the affiliating university. Students who satisfy these requirements are awarded a degree in their enrolled branch, while those who fail to do so are classified as non-completers. Because graduation criteria and assessment standards are identical across colleges and programs, variation in completion outcomes can be meaningfully compared across institutions. Together, standardized examination scores and graduation indicators provide complementary measures of academic achievement, capturing both continuous learning outcomes and extensive-margin progression through the degree.

3 Data and Research Design

3.1 Data Sources and Sample Construction

Our analysis combines multiple administrative datasets covering applicants to undergraduate engineering programs admitted through the centralized admissions system in a large Indian state. The core datasets include: (i) students' ranked preference lists and final program assignments from the centralized counseling process; (ii) entrance examination records from the state-level engineering admissions test, which report raw scores, ranks, and demographic information including gender, caste category, and family income; (iii) secondary school academic records, including 10th and 12th grade board examination scores; and (iv) semester-wise, subject-level academic outcomes during college, obtained from the affiliating university.

These datasets are linked at the individual student level using anonymized administrative identifiers. The admissions and preference data allow us to reconstruct students' choice sets, admission margins, and distance from program-specific cutoffs, which are central to our regression discontinuity design. Academic outcomes are measured using standardized semester examinations, which are comparable across colleges within a major due to the common curriculum and grading structure described in Section 2.2.

The final analysis sample is obtained after merging these datasets and applying a small number of sample restrictions. We restrict attention to students admitted through the centralized allocation mechanism, excluding management-quota and lateral-entry students who are not assigned based on entrance exam cutoffs. We further require a valid linkage between admissions records and semester examination outcomes. This results in some sample attrition, primarily because academic records are unavailable for a subset of colleges due to incomplete digitization, reporting delays, or institutional non-participation in specific years. As a result, our final sample includes students enrolled in colleges for which consistent academic outcome data are available. After imposing these restrictions, the main analysis sample consists of 55,250 students admitted to undergraduate engineering programs in the 2015 admissions cohort.[7]

3.2 Descriptive Statistics

Table 1 presents summary statistics for the main variables used in our analysis. Statistics are reported separately by caste category—General (GEN), Backward Caste (BC), and

[7]The sample construction proceeds as follows. We begin with 60,637 students who received a CET score and were allocated a program through the centralized mechanism. Excluding admissions through physically handicapped, sports, NCC, Muslim, and Christian quotas yields 57,525 students (95 percent). Restricting attention to six engineering majors yields 55,250 students (96 percent of regular-quota admissions and 91 percent of the initial total). Of these, 44,056 students (80 percent of 55,250) enroll in their allotted program.

Scheduled Caste / Scheduled Tribe (SC/ST)—and distinguish between the full sample of engineering majors and the Computer Science Engineering (CSE) major. Unless otherwise noted, we focus on patterns in the full sample and use the CSE columns to illustrate differences within a homogeneous, but large and highly demanded field of study.

Table 1: Summary Statistics

	Caste group							
	All		GEN		BC		SC/ST	
	All Majors (1)	CSE (2)	All Majors (3)	CSE (4)	All Majors (5)	CSE (6)	All Majors (7)	CSE (8)
Demographics								
Female (share)	0.40	0.50	0.46	0.53	0.38	0.50	0.35	0.43
Age	17.72	17.71	17.64	17.63	17.73	17.72	17.86	17.88
	(0.71)	(0.67)	(0.60)	(0.58)	(0.70)	(0.66)	(0.88)	(0.89)
	[17.20, 18.07]	[17.22, 18.04]	[17.19, 17.96]	[17.20, 17.95]	[17.20, 18.09]	[17.23, 18.06]	[17.21, 18.22]	[17.25, 18.24]
Academic prep.								
12th Marks	80.97	81.66	85.39	85.88	80.49	80.53	73.69	72.94
	(14.84)	(14.57)	(12.94)	(12.49)	(14.82)	(14.61)	(15.23)	(15.50)
	[70.67, 93.50]	[71.83, 93.83]	[78.67, 95.50]	[79.50, 95.50]	[70.17, 93.00]	[70.50, 92.83]	[61.17, 87.00]	[60.00, 86.83]
Entrance marks	34.53	35.26	38.36	38.65	33.84	34.08	28.97	29.17
	(10.28)	(10.75)	(11.95)	(12.12)	(8.90)	(9.21)	(6.62)	(7.00)
	[27.50, 38.12]	[28.12, 39.38]	[29.38, 44.38]	[29.38, 43.75]	[27.50, 36.88]	[28.12, 36.88]	[24.38, 31.87]	[24.38, 31.87]
Composite score	46.14	46.86	50.12	50.46	45.51	45.69	40.15	40.12
	(10.28)	(10.57)	(11.15)	(11.16)	(9.20)	(9.38)	(7.63)	(7.97)
	[38.81, 51.36]	[39.27, 52.18]	[41.93, 56.60]	[42.39, 56.26]	[39.01, 50.04]	[39.04, 50.13]	[34.55, 44.48]	[34.40, 44.33]
Costs								
Out-of-pocket	21491	21243	25683	24371	21142	19706	14334	17084
	(25836)	(25519)	(26892)	(26471)	(24014)	(23457)	(26753)	(27956)
	[0, 35000]	[0, 35000]	[0, 41900]	[0, 36300]	[0, 35000]	[0, 35000]	[0, 10000]	[0, 35000]
Outcomes								
Enrolled (share)	0.80	0.77	0.69	0.66	0.85	0.83	0.86	0.85
Number of students	55250	17940	19095	7317	26255	8124	9900	2499

Notes: The reported statistics represent means unless otherwise noted. GEN denotes the general caste students, BC denotes Backward Caste, and SC/ST combines Scheduled Castes and Scheduled Tribes. For each continuous variable, the rows show the mean in the first line, standard deviation in parentheses on the second line, and the 25th and 75th percentile in hard brackets on the third line. Columns split each caste group into the full sample (All Majors) and the Computer Science Engineering (CSE) major only.

The average student in the full sample is 17.7 years old at entry, and 40% of enrolled students are female. Academic preparation differs substantially across caste groups, reflecting both historical inequalities and the structure of affirmative action policies. General category students have higher mean 12th grade board examination scores and entrance examination marks than BC and SC/ST students, and these differences are reflected in composite admission scores that combine board examination performance and entrance test scores.

Enrollment take-up following program assignment is high across all groups, with approximately 80% of allotted students enrolling in their assigned program. Take-up rates are slightly higher among BC and SC/ST students, consistent with differences in outside options. Out-of-pocket tuition expenditures vary sharply by caste category, reflecting fee concessions and financial support under reservation policies.

To provide a concrete illustration within a single, large major, the table separately reports statistics for Computer Science Engineering (CSE), the most sought-after program in the system. Within CSE, the share of female students is 50%. Mean 12th grade board examination scores are 81.66 out of 100, and mean scores on the Centralized Entrance Test (CET) are 35.26 out of 100. Admissions are based on a composite score that weights board examination performance at 25% and CET scores at 75%. In terms of caste composition, approximately 41% of CSE students belong to the General category, 45% to the Backward Caste category, and 14% to the Scheduled Caste or Scheduled Tribe groups. Mean out-of-pocket tuition expenditure among CSE students is ₹ 21,243. Overall, about 77% of students take up their allotted seat and enroll in the program to which they are assigned through the centralized admissions mechanism.

While CSE is the most popular major, similar admission structures, cutoff patterns, and academic environments apply across other major engineering branches, including Electronics and Communication Engineering, Electrical Engineering, and Mechanical Engineering. As discussed in Section 2.3, semester examinations are standardized within each major and administered uniformly across all affiliated colleges. As a result, academic outcomes are directly comparable across institutions within a field of study, but not across different majors. Value-added comparisons are therefore interpretable only within majors. Appendix tables report descriptive statistics and results pooled across all majors.

3.3 Observational Value-Added Framework (OLS)

We begin by estimating program-level academic value-added using an observational Ordinary Least Squares (OLS) framework. This approach follows the value-added literature in education, which seeks to isolate institutional contributions to student outcomes after conditioning on observed student characteristics (Chetty et al., 2014; Koedel and Rockoff, 2015). Such models rely on having an adequate set of controls, and are widely used in empirical research and policy discussions, making them a natural benchmark against which to compare our regression discontinuity estimates.

We define program-level value-added as the average association between enrollment in a college-major program and subsequent academic outcomes, after accounting for differences in prior achievement and background characteristics. Formally, we estimate:

$$y_{ij} = \alpha_k + \sum_p \beta_p \cdot \mathbb{1}\{\text{prog}_{ij} = p\} + \boldsymbol{\delta}'\mathbf{X}_{ij} + \varepsilon_{ij}, \tag{1}$$

where y_{ij} denotes the cumulative standardized semester examination score for student i enrolled in program j. The indicator $\mathbb{1}\{\text{prog}_{ij} = p\}$ equals one if student i is enrolled in program p, with β_p capturing the estimated value-added associated with program p.

The vector $\mathbf{X}_{ij}$ includes baseline academic characteristics such as entrance examination scores and 12th grade board examination marks, along with demographic controls. The term α_k denotes fixed effects for caste category $k \in \{\text{GEN}, \text{BC}, \text{SC/ST}\}$, and ε_{ij} captures unobserved determinants of academic performance. Standard errors are clustered at the program level.

The coefficients β_p summarize differences in academic outcomes across programs, conditional on observed controls. Despite the inclusion of rich baseline covariates, these estimates may reflect both program characteristics and selection on unobserved factors—such as motivation, aspirations, family support, or non-cognitive skills—that influence both program choice and academic performance (Rothstein, 2009; Altonji et al., 2012). In certain settings, many of our controls (which include baseline academic achievement) may account for these individual characteristics, and help provide an unbiased estimate of value added (Chetty et al., 2014; Koedel and Rockoff, 2015). Yet, selection concerns may be especially relevant in selective higher-education systems with centralized admissions and affirmative action policies, where students may enter the same program through systematically different admission margins.

Accordingly, we interpret the OLS value-added estimates as important benchmarks for our RD estimates. OLS value-added measures are widely used in higher-education research and policy discussions (Cunha and Miller, 2014; Mountjoy and Hickman, 2021). They therefore serve as a natural benchmark for our RD estimates, clarifying when observational models approximate margin-specific causal effects and when they instead capture different parameters.

3.4 Regression Discontinuity Design (RDD)

To address the limitations of observational value-added estimates, we exploit discontinuities in program admission generated by the centralized assignment mechanism. Our approach builds on a large literature that uses admission cutoffs in schools and colleges to estimate the causal impact of access to higher-quality educational institutions (Pop-Eleches and Urquiola, 2013; Abdulkadiroğlu et al., 2014; Hoekstra, 2009; Zimmerman, 2014; Kumar, 2023). The key identifying assumption is that applicants just above and just below an admission cutoff are comparable in both observed and unobserved characteristics, so that discontinuities in outcomes can be attributed to program admission.

In our setting, admission to each college–major program p is determined by a standardized entrance examination score and a centralized preference-based allocation algorithm, with separate cutoffs for each affirmative-action category k. Because clearing a cutoff substantially increases—but does not perfectly determine—subsequent enrollment in the allotted program, we implement a *fuzzy* regression discontinuity

design.[8] Estimates are therefore interpreted as local average treatment effects (LATE) for students whose enrollment decisions are shifted by crossing the cutoff (Imbens and Angrist, 1994; Hahn et al., 2001; Imbens and Lemieux, 2008).

Running variable and treatment. Let i index students, p index college–major programs, and k index affirmative-action categories. Each admission cutoff is defined at the program–category level (p, k). We define the running variable as the student's entrance score centered at the relevant cutoff:

$$C_{ipk} = S_i - \overline{S}_{pk}, \tag{2}$$

where S_i denotes the composite admission score used in assignment and $\overline{S}_{pk}$ is the cutoff score for program p and category k. Define the cutoff-crossing indicator:

$$Z_{ipk} = \mathbb{1}\{C_{ipk} \geq 0\}. \tag{3}$$

Let D_{ipk} indicate enrollment in program p for student i at category-specific margin k.

First stage. Crossing an admission cutoff generates a discrete increase in the probability of enrollment. We estimate the first-stage relationship as:

$$D_{ipk} = \delta_{pk} + \rho Z_{ipk} + f(C_{ipk}) + \eta_{ipk}, \tag{4}$$

where δ_{pk} are program–category fixed effects and $f(\cdot)$ is a flexible function of the running variable estimated locally. The coefficient ρ captures the discontinuous change in enrollment probability at the cutoff and provides a test of instrument relevance.

Reduced form. For any outcome Y_{ipk}—including academic performance, graduation, peer quality, or within-cohort ordinal rank—we estimate the reduced-form effect of cutoff-crossing:

$$Y_{ipk} = \alpha_{pk} + \gamma Z_{ipk} + f(C_{ipk}) + \varepsilon_{ipk}, \tag{5}$$

where α_{pk} are program–category fixed effects. The coefficient γ captures the effect of clearing the cutoff on the outcome of interest.

Second stage and interpretation. We combine Equations (4) and (5) in a two-stage least squares framework, instrumenting enrollment with cutoff-crossing:

$$Y_{ipk} = \beta_{pk} + \lambda \widehat{D}_{ipk} + f(C_{ipk}) + u_{ipk}. \tag{6}$$

[8]Students admitted to a given program may choose not to enroll, for example by attending a private institution or enrolling in a different state or country.

Under standard fuzzy RD assumptions, λ identifies the local average treatment effect of marginal enrollment into program p at category-specific cutoff k, and is equal to the ratio of the reduced-form and first-stage discontinuities.

When Y_{ipk} is an academic outcome measured on a standardized scale, λ captures the local causal effect of marginal admission into the program—combining changes in peer composition, institutional environment, and ordinal rank—for students at the admission cutoff.

In practice, we stack observations across program–category cutoffs and estimate local linear specifications on either side of the threshold, allowing for different slopes above and below the cutoff. Our baseline implementation follows standard RD practice with bandwidth selection and inference procedures as in Imbens and Lemieux (2008) and Calonico et al. (2014). Standard errors are clustered at the program–category cutoff level.

4 Results

4.1 Observational Program Value-Added Using OLS

We begin by documenting patterns in program-level academic value-added using an observational Ordinary Least Squares (OLS) framework. As discussed in Section 3.3, we estimate program value-added by running Equation (1) separately for each program and recovering the associated program fixed effects. Under certain identifying assumptions—including that selection into programs is fully accounted for by observed controls, and functional form—these estimates can be interpreted as causal program impacts. More generally, however, OLS value-added provides a benchmark mirroring how value-added is typically measured and interpreted in the education literature. Under weaker assumptions, OLS value-added reflects a combination of program contributions and selection on both observed and unobserved determinants of achievement. For now, we treat these estimates primarily as descriptive objects, useful for characterizing heterogeneity and motivating the use of a research design that relaxes the required selection assumptions.

Figure 1 plots program-level OLS value-added estimates against average entrance examination scores, a proxy for program selectivity. Two patterns emerge. First, in Panel A, in the absence of controls, there is a strong positive relationship between program selectivity and estimated value-added: more selective programs exhibit substantially higher OLS value-added. Second, in Panel B, this relationship attenuates sharply once baseline academic achievement (proxied by lagged test scores) and demographic controls are included. The reduction in slope indicates that a large share of the raw selectivity–value-added gradient reflects that much of the initial association

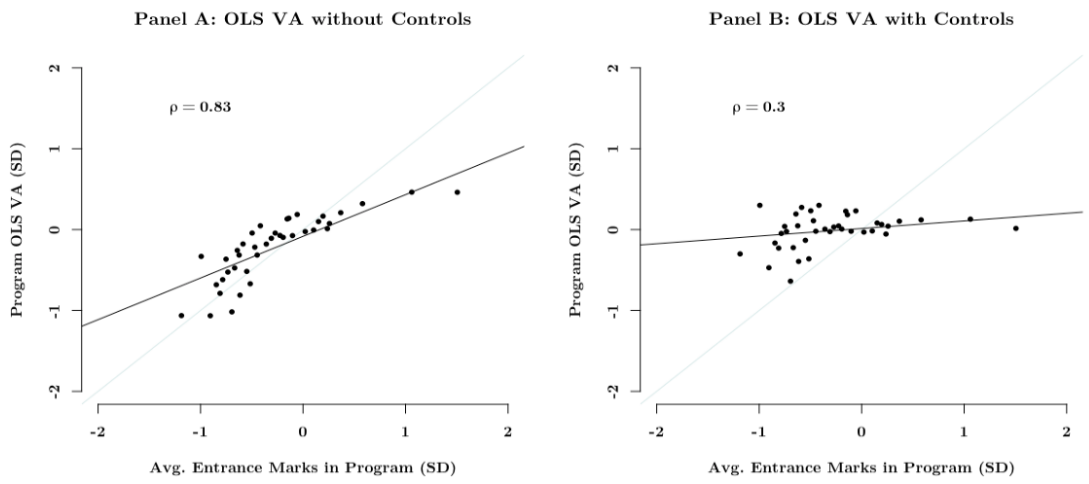

Figure 1: Program Selectivity and Observational Value-Added Estimates

Notes: This figure plots program-level observational value-added estimates against program selectivity. The horizontal axis reports the average entrance examination score of enrolled students in each college–major program, measured in standard deviations. The vertical axis reports the program-level OLS value-added estimate (program fixed effect), measured in standard deviations. Each point corresponds to a college–major program; the fitted line shows the best linear fit. Program value-added is estimated controlling for baseline academic measures (entrance exam score and 12th-grade board marks) and demographic characteristics, and includes caste-category fixed effects; standard errors are clustered at the program level.

comes from ignoring selection. Baseline test score controls may also help capture time-invariant student (unobserved) characteristics that are determinants of achievement, such as effort, motivation, or prior educational inputs. While the explicit inclusion of baseline test scores and demographic controls addresses some of these sources of bias, this correlation might not always be interpreted causally, as there might still be systematic sorting on ability levels across classrooms.

Figure 2 shows substantial dispersion in observational (OLS) value-added across programs. Even within a single major such as Computer Science Engineering (CSE), the interquartile range spans about 0.4 standard deviations, with the 25th percentile at −0.16 SD and the 75th percentile at 0.24 SD (Table B4). This degree of spread suggests meaningful heterogeneity in estimated program performance. At the same time, because these estimates are observational, the dispersion may reflect both differences in program contributions and differences in the types of students who sort into each program, including dimensions not fully captured by baseline measures. This motivates our regression discontinuity design, which leverages admission cutoffs to obtain causal estimates at the margin.

Figure 3 documents sharp heterogeneity in OLS value-added across caste groups. The distribution of OLS value-added for General-category students is centered close to zero and exhibits a relatively narrow spread. In contrast, the distributions for Backward Caste and Scheduled Caste/Scheduled Tribe students are shifted markedly leftward. At

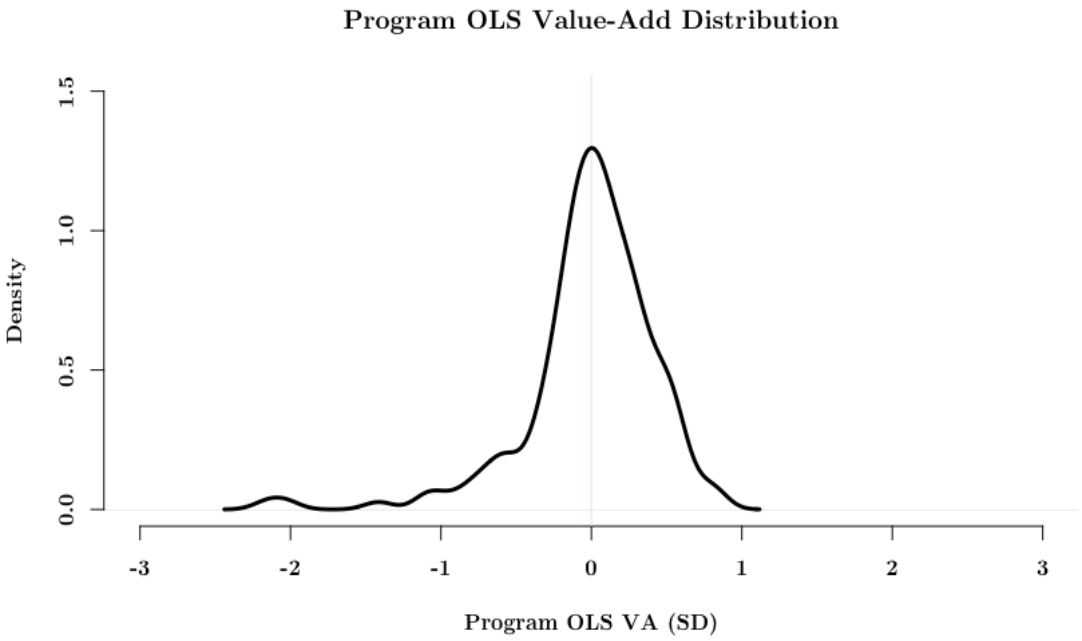

Figure 2: Distribution of Program OLS Value-Added

Notes: This figure plots the distribution of program-level observational value-added (OLS) estimates, measured in standard deviations. The horizontal axis reports the estimated program fixed effect from Equation (1), and the vertical axis reports the kernel density. Each program corresponds to a college–major unit. Estimates are observational and may reflect both program contributions and selection into programs. The sample is restricted to Computer Science Engineering (CSE) programs

the median, OLS value-added is approximately zero for General-category students, but close to -0.5 standard deviations for both BC and SC/ST students (Table B4). The lower tail of the distribution is also substantially thicker for disadvantaged groups, indicating that observational estimates assign systematically lower value-added to programs as experienced by these students.

Taken together, these patterns highlight a central puzzle. Observational OLS estimates suggest substantial heterogeneity in program performance and pronounced caste-based differences in estimated value-added. However, in a centralized admissions system with affirmative action, students enter the same program through distinct admission margins that may differ systematically in observed characteristics, like ability and unobserved inputs that directly affect academic achievement. As a result, OLS value-added may conflate true program effects with selection on dimensions not fully captured by baseline academic measures. In the next subsection, we exploit discontinuities in program admission to implement a regression discontinuity design, allowing us to estimate the causal effect of marginal admission into more selective programs.

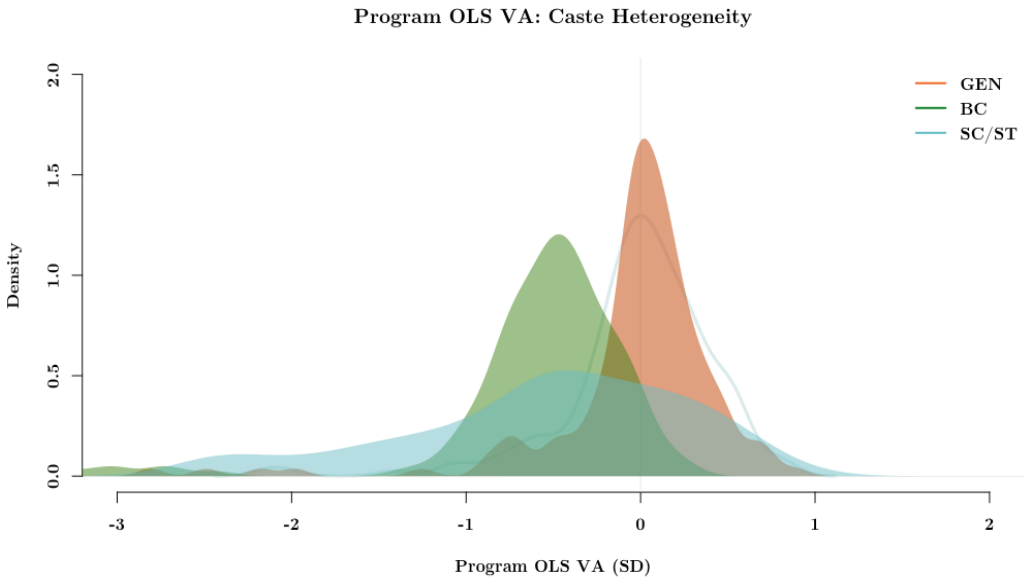

Figure 3: Heterogeneity in Observational Program Value-Added by Caste Group

Notes: This figure plots the distribution of program-level observational value-added (OLS) estimates separately for students in the General, Backward Caste (BC), and Scheduled Caste/Scheduled Tribe (SC/ST) categories. Program value-added is measured as the estimated program fixed effect from Equation (1), expressed in standard deviations of cumulative standardized examination scores. Each density reflects the distribution of program value-added as experienced by students in the corresponding caste group. Estimates control for baseline academic achievement and demographic characteristics and include caste-category fixed effects. As with all OLS-based measures, these estimates are descriptive and may reflect both program contributions and selection into programs.

4.2 Regression Discontinuity Estimates of Value Added

We now turn to regression discontinuity (RD) estimates of program value-added, which leverage the centralized admissions cutoffs described in Section 3.4. Admission to a program is determined by sharp score thresholds that vary by program and category, allowing us to compare students who narrowly clear a cutoff to those who narrowly miss it. These students are observationally similar in prior achievement and demographics, but may experience discrete differences in academic environments upon enrollment. We estimate a fuzzy RD design and report two-stage least squares (2SLS) local average treatment effects (LATEs), capturing the causal effect of enrolling in a more selective program for students at the margin of admission.

The first-stage relationship between admission eligibility and enrollment is strong: crossing the cutoff increases the probability of enrollment by approximately 0.88, as shown in Figure 4 and Table B1. While compliance is incomplete, the magnitude of the discontinuity indicates a highly relevant instrument for program enrollment. In other words, program allotment through the centralized admissions mechanism is a strong instrument for the probability of enrollment.

We next present standard diagnostic tests supporting the validity of the regression discontinuity design. Figure 5 shows no evidence of sorting around admission cutoffs,

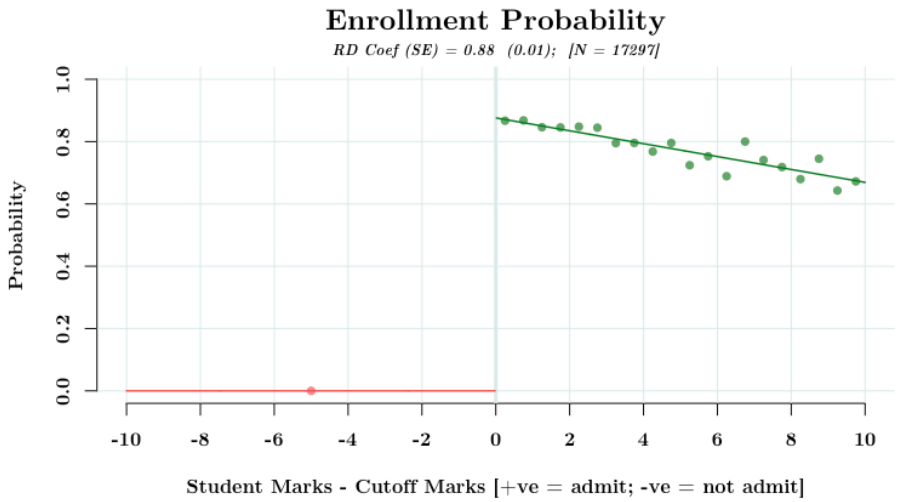

Figure 4: First Stage: Probability of Enrollment (CSE Only)

Notes: The horizontal axis reports the running variable, defined as the entrance examination score centered at the program-specific admission cutoff. The vertical axis reports the probability of enrollment in the allotted program. Local polynomial fits are estimated using the preferred bandwidth selected by data-driven bandwidth choice procedures. A sharp increase at the cutoff indicates a strong first-stage relationship between admission eligibility and enrollment.

while Figure 6 and Table B2 show that predetermined characteristics evolve smoothly through the cutoff. Together, these results support the identifying assumption that students just above and below the cutoff are comparable.

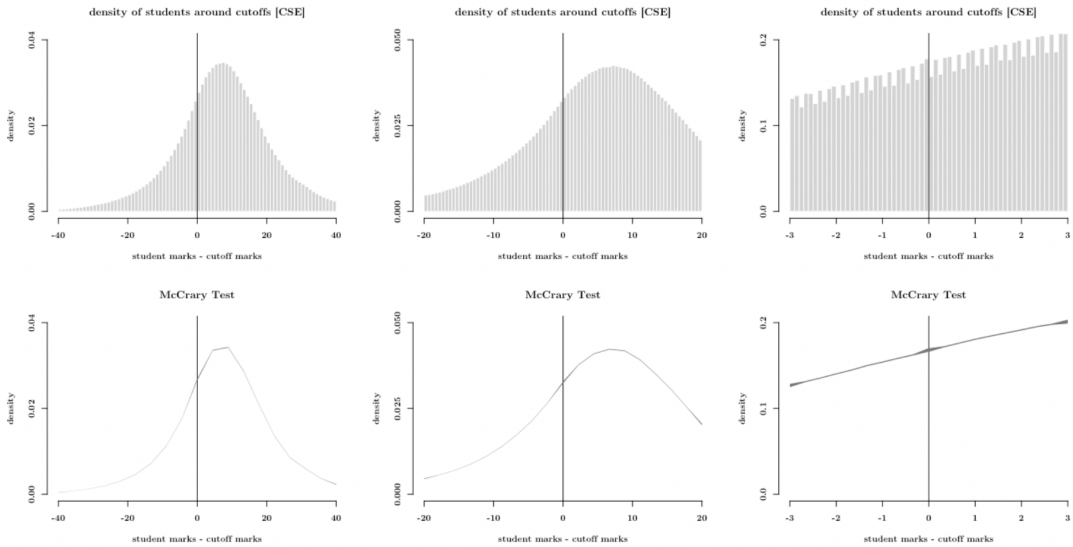

Figure 5: McCrary Density Tests Around Admission Cutoffs

Notes: This figure presents McCrary density tests of entrance examination scores around program admission cutoffs. The horizontal axis reports the running variable, defined as the entrance examination score centered at the cutoff. Panels display results across alternative bandwidths. No statistically significant discontinuity in density is detected at the cutoff, indicating no evidence of manipulation of the running variable.

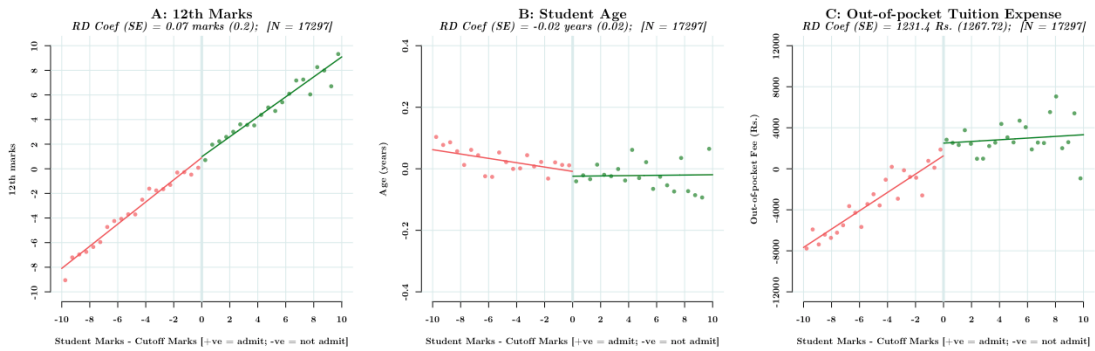

Figure 6: Covariate Smoothness at Admission Cutoffs

Notes: This figure examines the smoothness of predetermined covariates around program admission cutoffs. The horizontal axis reports the running variable, defined as the entrance examination score centered at the program-specific cutoff. Panels plot residualized 12th-grade marks, student age, and out-of-pocket tuition expenditure. Local polynomial fits are estimated on either side of the cutoff using the preferred bandwidth. No discontinuities are visible at the cutoff, supporting the assumption that students just above and below the cutoff are comparable in predetermined characteristics.

Figure 7 shows clear reduced-form discontinuities in academic outcomes (academic achievement and graduation probability) at admission cutoffs. Marginal admission increases total exam scores by approximately 0.11 standard deviations and raises graduation probability by 3–4 percentage points (Tables 2 and B5). These effects are stable across bandwidths and persist across years of study.

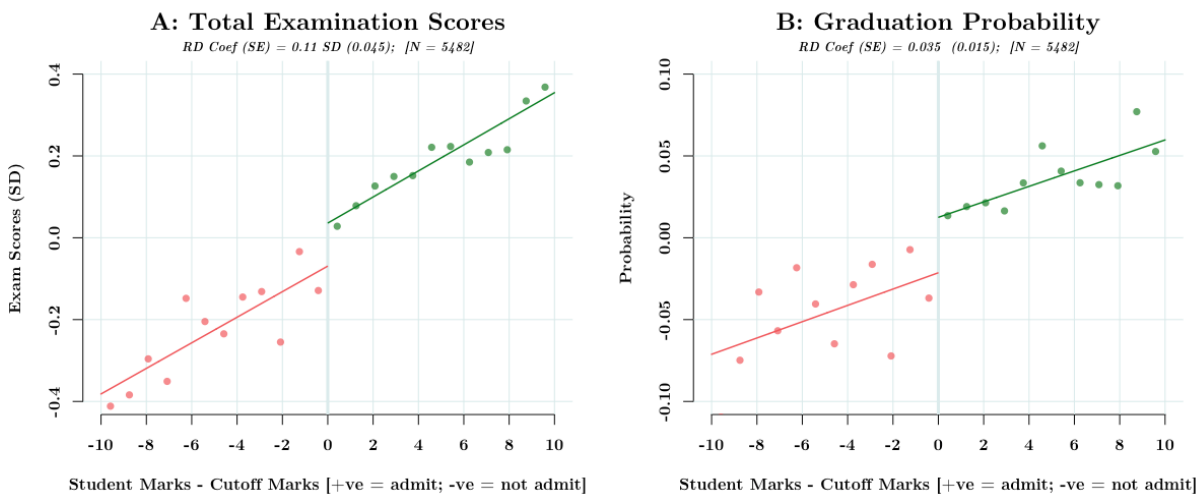

Figure 7: Reduced Form: Academic Outcomes at Admission Cutoffs (CSE Only)

Notes: The horizontal axis reports the running variable, defined as the entrance examination score centered at the program-specific admission cutoff. Panel A plots residualized total academic marks (standard deviations), and Panel B plots the residualized probability of graduating. Local polynomial fits are estimated on either side of the cutoff using the preferred bandwidth selected by data-driven bandwidth choice procedures. Discontinuities at the cutoff represent reduced-form effects of admission eligibility on academic outcomes.

While our previous results in this section focused on the *aggregate* impact of clear-

Table 2: Regression Discontinuity Estimates of Academic Outcomes

	±6		±10		±14		±18	
	All Majors (1)	CSE (2)	All Majors (3)	CSE (4)	All Majors (5)	CSE (6)	All Majors (7)	CSE (8)
Total Marks (SD)	0.113***	0.116**	0.115***	0.110**	0.108***	0.106**	0.112***	0.112***
	(0.039)	(0.057)	(0.030)	(0.045)	(0.029)	(0.043)	(0.028)	(0.039)
$\mathbb{P}$(Graduation = 1)	0.040***	0.037**	0.034***	0.035**	0.031***	0.030**	0.033***	0.031**
	(0.013)	(0.018)	(0.010)	(0.015)	(0.009)	(0.013)	(0.009)	(0.013)
Year 2 Marks (SD)	0.079**	0.098*	0.096***	0.088*	0.091***	0.083*	0.093***	0.092**
	(0.039)	(0.056)	(0.030)	(0.044)	(0.030)	(0.043)	(0.029)	(0.039)
Year 3 Marks (SD)	0.115***	0.122**	0.119***	0.124***	0.109***	0.118***	0.114***	0.121***
	(0.040)	(0.055)	(0.030)	(0.043)	(0.029)	(0.043)	(0.028)	(0.040)
Year 4 Marks (SD)	0.123***	0.124**	0.108***	0.112**	0.098***	0.097**	0.103***	0.103**
	(0.039)	(0.056)	(0.030)	(0.045)	(0.028)	(0.041)	(0.028)	(0.039)
N	9,233	4,319	11,964	5,482	13,047	5,919	13,516	6,109

Notes: Each cell reports a 2SLS estimate from a fuzzy regression discontinuity design, where enrollment in the allotted program is instrumented by admission eligibility. Results are shown for the full sample of majors and separately for Computer Science and Engineering (CSE), across alternative entrance-exam score bandwidths around the admission cutoff. Outcomes are standardized within cohort. Standard errors (in parentheses) are clustered at the program level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

ing an admissions cutoff on outcomes of interest, we now estimate program-specific RD value-added, for programs within the CSE major. The estimating equation for individual RD program value added is given by Equation (7).

$$y_{ij} = \alpha_j^{prog} + \sum_{l=1}^{J} \beta_l^D \times D_{ij} \times \mathbb{1}[l = j] + \beta^X X_{ij} + \beta^{DX} D_{ij} X_{ij} + \epsilon_{ij} \tag{7}$$

Figure 8 plots the distribution of β_j^D, i.e., the RD VAM estimated for each program individually.[9] The median program VA is 0.14 SD, and movement from the 25th to the 75th percentile of the program VA distribution corresponds to an increase of approximately 0.61 SD (Table B4). We also note that the distribution has a long right-hand tail, with some programs having a much higher positive value-added relative to the modal program.

We next examine heterogeneity by caste group. Figure 9 shows that RD value-added estimates are positive across all caste categories. Median effects are similar in magnitude for General and Backward Caste students, while Scheduled Caste and Scheduled Tribe students also experience positive gains, albeit with somewhat greater dispersion. Unlike the OLS results in Section 4.1, the RD estimates do not exhibit large caste-specific penalties. Using an RD approach, we find that comparing the outcomes of students in the neighborhood of admission cutoffs reveals that the beneficiaries of

[9]The program RD VAM distribution in Figure 8 corresponds to a specification where we exclude caste fixed effects. In Figure B1, we show that the distribution does not exhibit any noteworthy changes upon the inclusion of caste fixed effects.

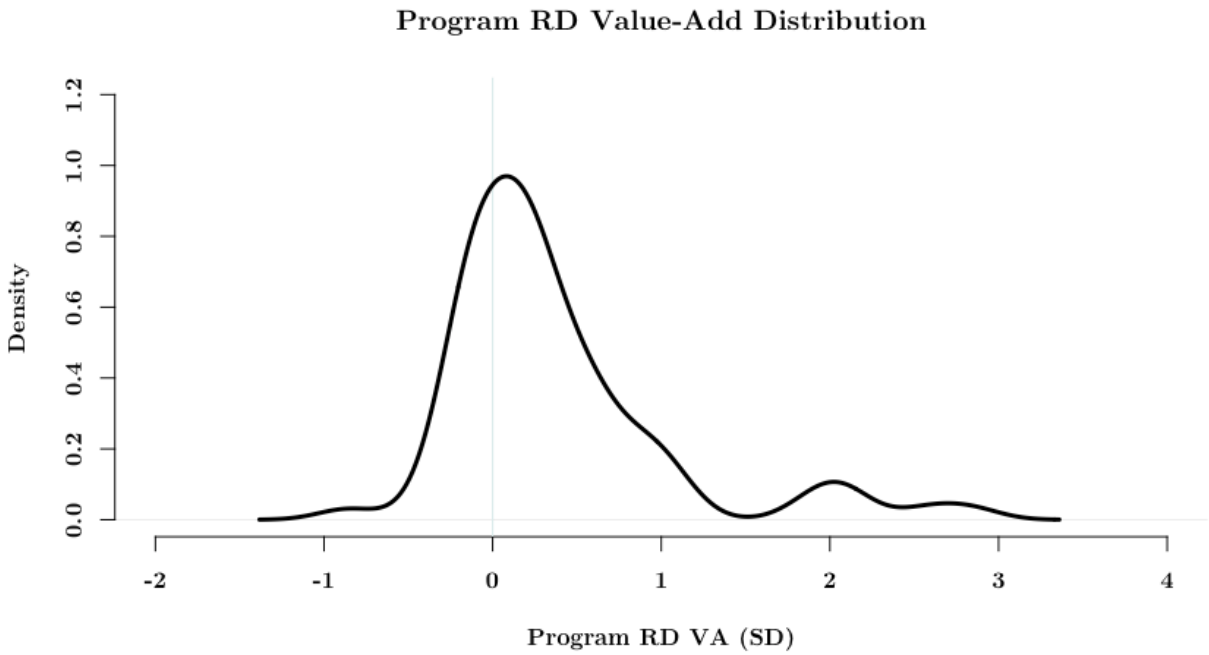

Figure 8: Distribution of Program RD Value-Added Estimates

Notes: This figure plots the distribution of program-level RD value-added estimates, measured in standard deviation units. The horizontal axis reports the estimated effect of marginal admission on total academic marks for students at the cutoff, and the vertical axis reports the kernel density. Each observation corresponds to a college–major program.

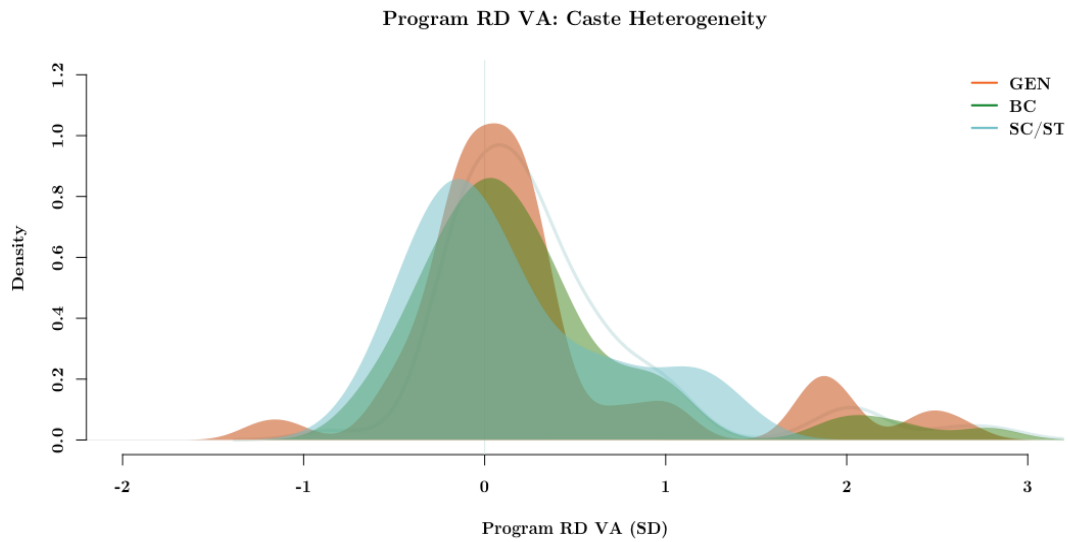

Figure 9: Heterogeneity in RD Value-Added by Caste (CSE Only)

Notes: This figure plots kernel density estimates of program-level RD value-added separately by caste group. The horizontal axis reports the RD value-added estimate (standard deviations), measured as the effect of marginal admission on total academic marks. Density lines correspond to General, Backward Caste, and Scheduled Caste/Scheduled Tribe students.

affirmative action experience, on average, higher value-added than suggested by OLS.

Taken together, these findings establish that access to more selective programs

causally improves academic outcomes for students at the margin of admission, particularly for AA (Affirmative Action) students. In the next subsection, we directly compare OLS and RD estimates to understand when and why observational value-added measures succeed or fail in ranking programs.

4.3 Comparing OLS and RD Program Value-Added

We now compare observational and causal estimates of program-level value-added to assess when OLS estimates align with regression discontinuity (RD) estimates and when they diverge. While OLS value-added measures conditional associations between program enrollment and academic outcomes, RD value-added identifies the LATE of marginal admission into a program. Comparing the two is informative for understanding the limitations of various models and for evaluating the reliability of value-added metrics used in the education literature for measuring institutional and teacher performance.

Table 3: OLS and RD Specifications

	OLS	**RD**
Outcome	Total exam scores (s.d.)	Total exam scores (s.d.)
Fixed effects	Actual program; caste	Program waitlist
Controls	High school score (s.d.); CET score (s.d.)	Running variable; running variable $\times\ \mathbb{1}\{X_{ijk} \geq 0\}$
Standard errors	Clustered at actual program	Clustered at program waitlist

Notes: This table summarizes the regression specifications used to construct observational (OLS) and regression-discontinuity (RD) value-added measures. Guided by the value-added literature (Chetty et al. 2014; Koedel and Rockoff 2015) and the patterns in Figure 1, the preferred OLS specification includes caste fixed effects and lagged achievement controls (standardized high-school and college entrance examination scores). The preferred RD specification uses a fuzzy RD design within a specified bandwidth around program admission cutoffs and includes the running variable and its interaction with the admission-eligibility indicator. Consistent with Figure 8, the preferred RD specification omits caste fixed effects because RD value-added estimates are not sensitive to the inclusion of additional controls in the local neighborhood around the cutoff.

While Sections 4.1 and 4.2 examined OLS and RD value-added separately, Figure 10 directly overlays the two distributions. Panel A shows that the distribution of OLS program value-added is substantially more dispersed than the corresponding RD distribution. In particular, the OLS distribution exhibits heavier left tails, with many programs displaying negative estimated value-added. In contrast, the RD distribution is more tightly concentrated and has a heavier right tail. Consistent with this pattern, the median program value-added under RD is approximately 0.13 standard deviations, compared to a median of roughly 0.04 standard deviations under OLS (Table B4), suggesting that observational estimates may understate academic gains from program access for a nontrivial subset of programs.

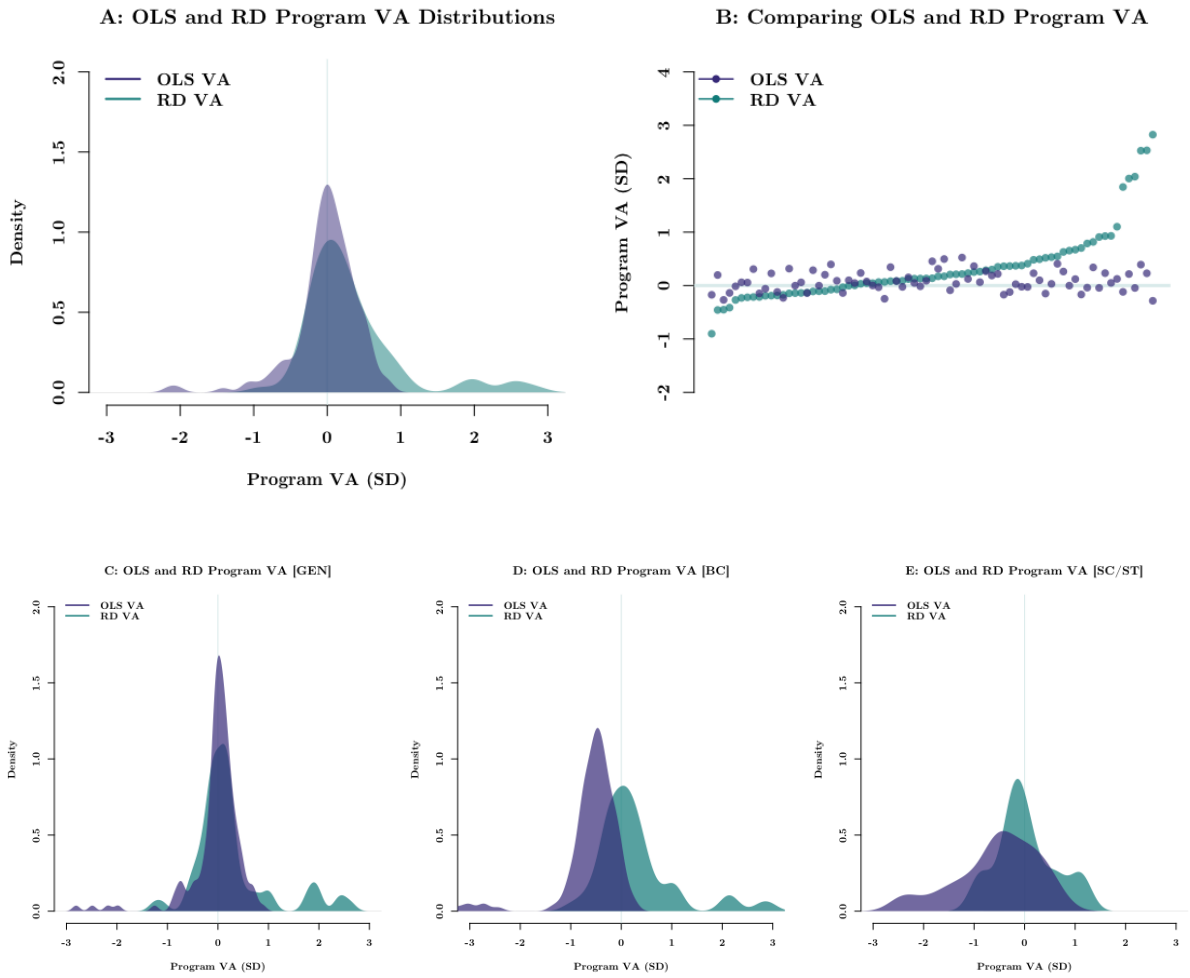

Figure 10: Comparison of OLS and RD Program Value Added

Notes: Panels A and B compare program-level value-added measured using ordinary least squares (OLS) and regression discontinuity (RD) designs. Panel A plots kernel density estimates of program value-added (in standard deviation units). Panel B plots the pairwise relationship between OLS and RD value-added at the program level. Panels C, D, and E report program-level value-added distributions separately by caste group: General (GEN), Backward Caste (BC), and Scheduled Caste/Scheduled Tribe (SC/ST), respectively. In all panels, blue curves denote OLS estimates and purple curves denote RD estimates. The horizontal axis reports program value-added (standard deviations), and the vertical axis reports the corresponding density.

Panel B of Figure 10 plots program-level OLS and RD value-added estimates against each other. While the two measures are positively correlated, the relationship is far from one-to-one. Programs with high OLS value-added tend to have positive RD estimates, but the magnitudes often differ substantially. More importantly, several programs that appear to have low or even negative value-added under OLS, exhibit positive value-added when we apply an RD approach.

Panels C, D, and E of Figure 10 summarize the relationship between OLS and RD estimates separately for each major caste group—General (GEN), Backward Caste (BC), and Scheduled Caste/Scheduled Tribe (SC/ST), respectively. Disaggregating the

OLS–RD comparison by caste group reveals important heterogeneity masked in the aggregate results. In Panel C, for GEN students, OLS and RD estimates align closely, with distributions overlapping substantially, though the RD distribution exhibits a somewhat heavier right-hand tail. In Panel D, for BC students, the OLS distribution shows substantial mass below zero and suggests that most programs offer negative gains. In contrast, the RD value-added distribution shifts rightward, centered closer to zero with more mass in the right tail. This implies that for marginal BC admits, RD indicates non-negative average program effects with a subset of programs delivering sizable gains, whereas OLS suggests negative effects. In Panel E, for SC/ST students, the OLS curve is wider and left-shifted, while the RD curve shifts mass toward zero and positive values and exhibits a pronounced right tail beyond one standard deviation. Thus, for marginal SC/ST admits, RD indicates that many programs yield zero to positive gains, with some large gains, whereas OLS would imply broadly negative effects. Table B4 presents additional summary statistics quantifying the percentiles of the value-added distributions for the overall sample and for each caste group.

Taken together, these patterns suggest that OLS and RD value-added seem reasonably similar when admission margins are similar, but differ precisely in settings where admission induces changes in students' academic environment. Because RD estimates capture the effect of marginal program access while OLS may conflate institutional effects with selection, observational models might misattribute these forces to program quality itself. Yet, the RD and OLS also estimate different parameters: the RD measures the effect for individuals around the cutoff, rather than the average student. Students who barely qualify for more desirable programs are often the lowest-ranked within those classrooms, whereas they are compared to the top-ranked students in less desirable classrooms. This comparison may thus be confounded by *rank effects*, which we need to isolate.

The advantage of our setting is that having multiple admissions cutoffs within a classroom allows us to develop new methods to estimate rank effects, and isolate them from overall value added for the first time. In the next subsection, we use the RD design to characterize how marginal admission reshapes students' peer quality and relative rank at different cutoffs. This lays the groundwork for us to explore the mechanisms underlying program value-added in Section 5.

4.4 Peer Quality and Ordinal Rank Effects

In this subsection, we establish two empirical facts that are central to interpreting the RD estimand. First, crossing an admission cutoff and entering a preferred program reshapes students' academic environment along two interlinked dimensions. Specifically, compliers who enroll in a more selective program experience an improvement

in peer quality and a deterioration in ordinal rank within the admitted cohort. We document the magnitude of these discontinuities at different cutoffs, and interpret RD value-added as the net effect of these shifts in peer quality and rank. Second, the magnitudes of these changes vary across category-specific admission cutoffs within the same program, owing to AA policies. This heterogeneity generates the variation required to separately identify the contribution of ordinal rank to program-level value-added.

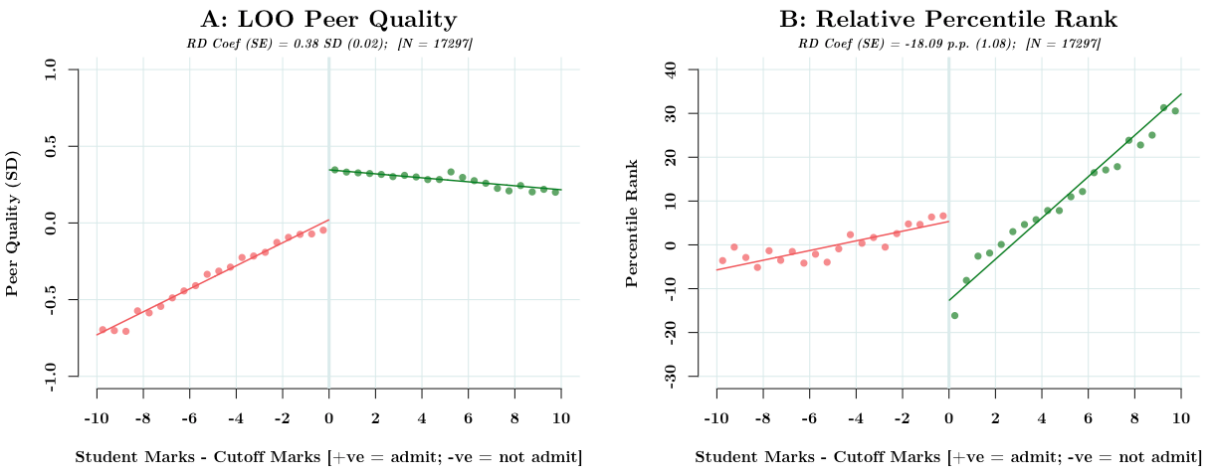

Figure 11: Peer Quality and Relative Rank at Admission Cutoffs

Notes: This figure illustrates reduced-form regression discontinuity estimates of peer quality and relative rank around program admission cutoffs. The horizontal axis reports the running variable, defined as the entrance examination score centered at the program-specific admission cutoff. Panel A plots residualized leave-one-out average peer entrance examination scores within a program, measured in standard deviation units. Panel B plots the residualized relative percentile rank of a student within the admitted cohort. Local polynomial fits are estimated on either side of the cutoff using the preferred bandwidth selected by data-driven bandwidth choice procedures. Discontinuities at the cutoff indicate changes in peer composition and within-cohort rank induced by marginal admission.

Figure 11 illustrates the discontinuities in peer quality and relative rank around program admission cutoffs (pooling across caste-based admission categories). Panel A shows that clearing a cutoff leads to a large and statistically significant increase in peer quality. On average, marginally admitted students are exposed to peers whose entrance exam scores are approximately 0.39 standard deviations higher than those of peers in their fallback programs. This pattern reflects a substantial upgrade in the peer environment associated with selective program access. Similar or larger magnitudes are observed when attention is restricted to affiliated programs (Table B3).

Panel B of Figure 11 shows that this improvement in peer quality is accompanied by a significant decline in relative rank. Marginally admitted students experience an average drop of roughly 18 percentile points in their position within the classroom relative to students who just miss the cutoff and enroll in a less selective program. Importantly, this decline is mechanical: marginally admitted students move into a more selective cohort, where they are ranked lower than they would have been in their

fallback program.

The primary takeaway from Figure 11 is that marginal admission simultaneously raises peer quality and lowers students' ordinal rank within the destination cohort. Yet, peer quality and relative ranks may have ambiguous effects. Lower relative rank may depress achievement through behavioral or social channels (Fabregas, 2023). While most work documents positive peer effects, in certain settings, there may be negative effects of high-achieving peers as well (Busso and Frisancho, 2021). The RD estimand, therefore, aggregates these potentially opposing components into a single measure of program value-added.

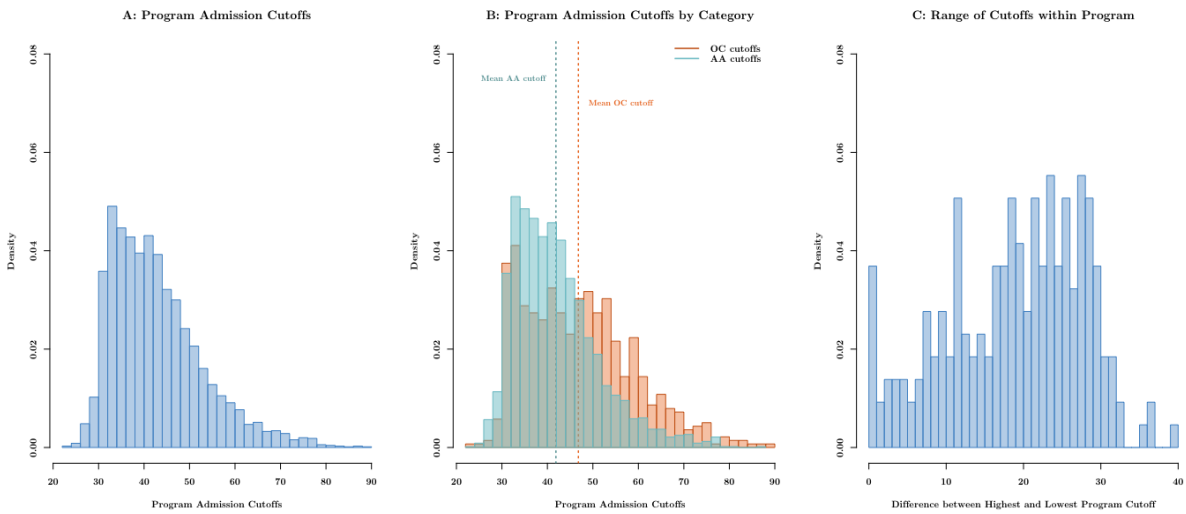

Figure 12: Distribution of Program Admission Cutoffs

Notes: Panel A plots the distribution of program-specific admission cutoffs across all seat categories. The horizontal axis reports the entrance examination score required for admission into a given program, and the vertical axis reports the corresponding density. Panel B plots the distribution of admission cutoffs separately by seat category, distinguishing Open Category (OC) and Affirmative Action (AA) cutoffs. Vertical dashed lines indicate category-specific mean cutoffs across all programs. Differences across distributions reflect variation in admission thresholds across seat categories within the same programs. Panel C plots the distribution of the range of cutoffs within a program.

The source of cross-cutoff variation in induced rank changes is institutional. Because admission to each program is governed by separate category-specific seat allocations, Open Category (OC) and Affirmative Action (AA) applicants face distinct admission thresholds within the same program. Figure 12 documents this feature of the admission system. Panel A plots the overall distribution of program admission cutoffs, while Panel B shows that OC and AA cutoffs differ systematically. In particular, OC cutoffs tend to be higher on average, reflecting more intense competition for unreserved seats. Panel C demonstrates that there is significant variation in admission cutoffs within a program, i.e. the difference between the highest and lowest cutoff within a program is not trivial.

This dispersion in admission thresholds implies that students admitted through

different seat categories enter the same destination program from different points in the underlying score distribution. As a result, the induced change in ordinal rank upon admission need not be identical across cutoff types. Variation in admission thresholds across categories, therefore, translates into variation in relative rank changes within the same program.

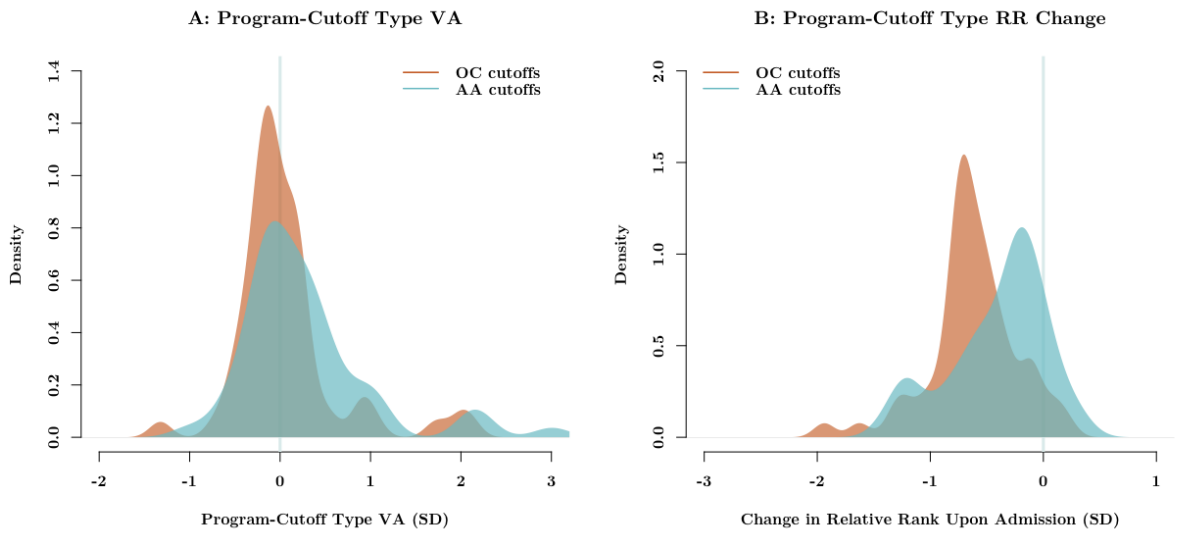

Figure 13: Program-Cutoff Heterogeneity in Value Added and Relative Rank

Notes: This figure compares distributions of regression-discontinuity estimates across different program admission margins. Panel A plots the distribution of program–cutoff-level value-added estimates for Open Category (OC) and Affirmative Action (AA) cutoffs. Value added is measured as the effect of marginal admission on total academic marks, in standard deviation units. Panel B plots the corresponding distributions of changes in students' relative percentile rank, in standard deviation units, within the admitted cohort at OC and AA cutoffs. In both panels, the vertical axis reports the kernel density. Differences across distributions reflect heterogeneity in academic gains and rank effects across admission margins within the same programs.

Identification of the relative-rank component of program value-added requires that induced rank changes and program value-added differ across category-specific admission margins within programs. If program value-added and rank changes moved in fixed proportion across cutoff types, the two components would be observationally indistinguishable. What matters, therefore, is whether the distributions of RD value-added and induced rank changes diverge across cutoffs within a program. To the extent that these distributions do not perfectly align, they generate independent variation in relative rank holding the destination program fixed, which allows us to isolate its contribution to overall program value-added. We develop the underlying conceptual framework in Section 5.

We now demonstrate empirically that this cross-cutoff heterogeneity generates variation in both RD value-added and induced rank changes. Figure 13 presents the cutoff-type specific density distributions of program VA and relative rank effects as measured by the RD approach. Panel A plots the distribution of program RD value-

added separately for OC and AA cutoffs. Movement from the 25th to the 75th percentile of the distribution corresponds to gains of 0.4 and 0.66 standard deviations for OC and AA students, respectively. Panel B plots the corresponding distributions of induced changes in relative rank, showing that the median marginally admitted OC and AA students experience, on average, a -0.63 and -0.28 standard deviation unit decline in relative rank as compared to students who just miss the program admission cutoffs (Table B6). The lack of perfect overlap between OC and AA value-added and relative rank declines introduces idiosyncratic variation that we exploit to isolate the component of program value-added that is attributable to changes in ordinal rank caused by crossing an admissions cutoff.

We argue that RD value-added estimates for a program recover the net effect of institutional quality and changes in a student's academic environment when they are admitted to more selective programs. We leverage the *plausibly* exogenous variation in within-program relative rank effects, facilitated by multiple admission cutoffs for a single program, to isolate the contribution of changes in ordinal rank to overall program value-added. In the next section, we formalize this intuition by developing a framework that decomposes program value-added into relative rank and other academic quality components. We show these mechanisms can be separately identified and quantified.

5 Conceptual Framework for Value-Added Mechanisms

The regression discontinuity (RD) design described in Section 3.4 identifies the effect of enrolling in a more selective program for students at the margin of admission. We refer to this as program-level value-added. Crossing an admission cutoff simultaneously changes several features of a student's academic environment. The advantage of our setting is that we see multiple admissions cutoffs within each program, allowing us to separate out the channels that affect student achievement. This section formalizes these channels and clarifies which components are identified by variation in our setting.

5.1 Conceptual Model of Value-Added

Consider the academic outcome of student i enrolled in program j:

$$Y_{ij} = \gamma^C C_j + \gamma^P P_j + \gamma^{RR} RR_{ij} + A_i + \varepsilon_{ij}, \tag{8}$$

where C_j captures college-level inputs (faculty, infrastructure, pedagogy), P_j denotes peer quality in program j, RR_{ij} is student i's ordinal rank within the program, A_i captures student ability and other time-invariant determinants of achievement, and ε_{ij} is an idiosyncratic error term.

Equation (8) highlights three channels through which program assignment may affect outcomes: institutional inputs, peer composition, and relative rank. The RD design identifies the combined effect of the discrete changes in these components induced by crossing an admission cutoff. Because institutional inputs and peer composition shift jointly at the program level, they cannot be separately disentangled without additional sources of variation.

For identification, we therefore aggregate institutional inputs and peer quality into a single program-level academic environment index,

$$Q_j \equiv \gamma^C C_j + \gamma^P P_j.$$

Substituting into Equation (8) yields:

$$Y_{ij} = \gamma^Q Q_j + \gamma^{RR} RR_{ij} + A_i + \varepsilon_{ij}, \tag{9}$$

where γ^Q captures the reduced-form effect of exposure to the academic environment at a more selective program. This is a combination of college inputs (like infrastructure and teaching staff), and as we have seen earlier, a higher-quality peer group. γ^{RR} captures the effect of changes in ordinal rank on the overall program value-added.

The sign and magnitude of γ^{RR} are empirical questions. If lower relative standing discourages effort or affects academic self-perception, γ^{RR} may be negative (Fabregas, 2023). The RD design in our setting provides variation that allows us to test this directly.

5.2 Program Value-Added at Multiple Admission Margins

Admission to a more selective program induces discrete changes in both the academic environment and ordinal rank. Let $\gamma_{j,k}$ denote the RD estimand associated with crossing admission cutoff-type k for program j. Using Equation (9), the cutoff-specific RD estimand can be written as:

$$\gamma_{j,k} = \gamma^Q \Delta Q_{j,k} + \gamma^{RR} \Delta RR_{j,k}, \tag{10}$$

where $\Delta Q_{j,k}$ and $\Delta RR_{j,k}$ denote the changes in academic environment and relative rank induced by admission at cutoff k (see Appendix A for the complete derivation).

Equation (10) makes clear that RD program value-added aggregates two components: changes in the academic environment and declines in ordinal rank. The total RD effect at each margin reflects the net contribution of these forces.

A key feature of our setting is that each college–major program admits students through multiple category-specific cutoffs generated by affirmative action policies. Students admitted through different cutoffs enroll in the same destination program and

therefore face identical institutional inputs, curriculum, faculty, and peer composition. However, because cutoff locations differ across categories, the implied shift in ordinal rank upon admission differs across margins.

Consider two admission margins within program j, corresponding to an open-category cutoff (OC) and an affirmative-action cutoff (AA). For each cutoff $k \in \{OC, AA\}$, the RD estimand is given by Equation (10). Taking the difference across cutoffs within the same program yields:

$$\gamma_{j,OC} - \gamma_{j,AA} = \gamma^{RR} \left(\Delta RR_{j,OC} - \Delta RR_{j,AA} \right), \tag{11}$$

provided that the change in academic environment is common across margins within program j. This condition requires that students who narrowly miss the OC and AA cutoffs enroll in fallback programs with comparable institutional inputs and peer composition. We provide supporting evidence for this assumption in Section 4.4 by comparing peer-quality shifts across margins.

Under this condition, cross-cutoff variation within the same program differences out the academic-environment component and isolates the ordinal-rank channel. Differences in RD value-added across admission margins therefore identify γ^{RR}.

5.3 Estimating the Ordinal Rank Effect

We begin by estimating the causal effect of ordinal rank on academic outcomes, holding the academic environment fixed. As shown in Section 4.4, marginal admission to a more selective program raises peer quality but simultaneously lowers a student's relative standing within the cohort. The multi-cutoff structure described in Section 5.2 allows us to isolate this ordinal-rank channel using within-program variation across admission margins.

Specifically, we estimate γ^{RR} by exploiting differences in value-added across category-specific cutoffs within the same college–major program, with program fixed effects absorbing common changes in institutional inputs and peer composition. This strategy identifies the effect of changes in relative standing under the assumption that fallback options are comparable across admission margins. The estimated rank effect is negative and statistically significant. A decline of 1 SD in relative rank reduces academic value-added by approximately γ^{RR} standard deviations.

To illustrate the quantitative importance of this channel, Figure 14 plots the distribution of total RD value-added and its decomposition into ordinal-rank and residual academic-environment components. The horizontal axis reports value-added in standard deviation units of total academic marks, and the vertical axis reports kernel density. The rank component, $\gamma^{RR}\Delta RR_{j,k}$, is centered below zero, indicating that declines in relative standing systematically reduce performance at selective admission margins.

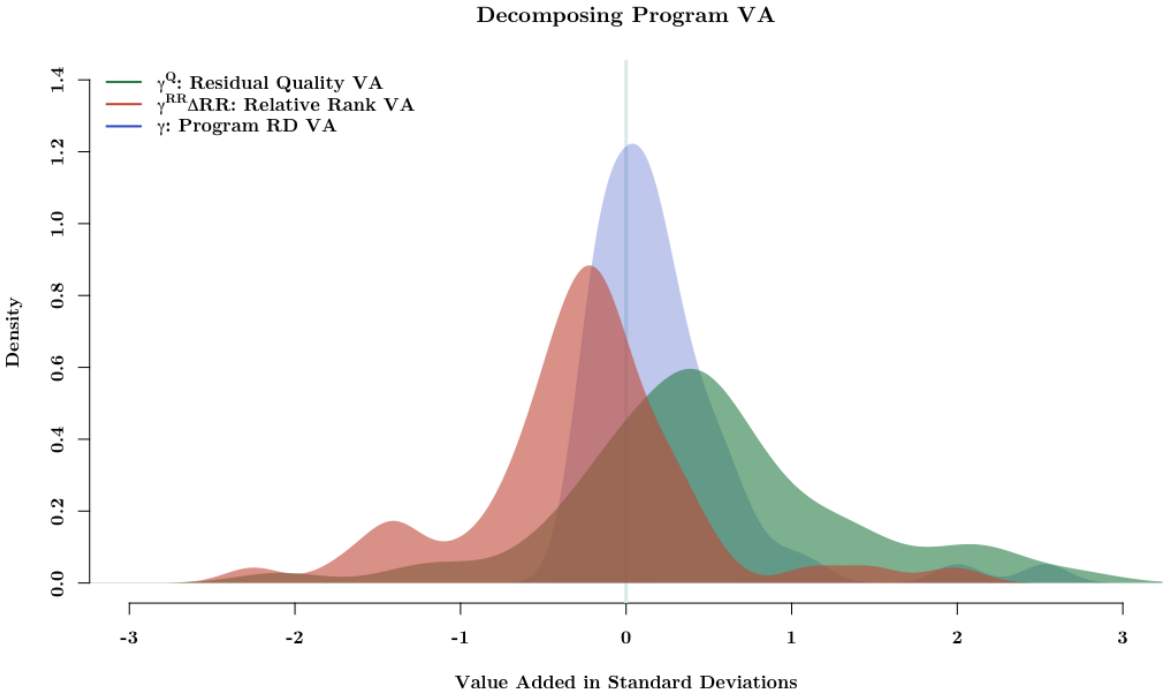

Figure 14: Decomposition of Program-Level RD Value-Added

Notes: This figure plots the distribution of regression discontinuity (RD) program value-added estimates and their decomposition into ordinal-rank and residual academic-environment components. The horizontal axis reports value-added measured in standard deviation units of total academic marks, and the vertical axis reports kernel density. The total RD value-added for program–cutoff cell (j, k) is denoted $\gamma_{j,k}$ and is decomposed as $\gamma_{j,k} = \gamma^{RR}\Delta RR_{j,k} + \gamma^{Q}_{j,k}$, where $\gamma^{RR}\Delta RR_{j,k}$ captures the ordinal-rank component and $\gamma^{Q}_{j,k}$ is defined in Equation (12). Each observation corresponds to a program–cutoff cell.

Table 4: Percentiles of Program Value-Added Components

Value-Added Component	25th pct.	50th pct.	75th pct.	IQR
$\gamma_{j,k}$ (Program RD VA)	-0.106	0.124	0.363	0.468
$\gamma^{RR}\Delta RR_{j,k}$ (Relative Rank Component)	-0.646	-0.244	-0.025	0.621
$\gamma^{Q}_{j,k}$ (Residual Academic Environment Component)	0.046	0.441	0.973	0.927

Notes: This table reports selected percentiles of the decomposition of program-level regression discontinuity (RD) value-added, expressed in standard deviation units. $\gamma_{j,k}$ denotes the RD value-added at admission cutoff k for program j. The relative-rank component equals $\gamma^{RR}\Delta RR_{j,k}$, and the residual academic-environment component is defined as $\gamma^{Q}_{j,k} = \gamma_{j,k} - \gamma^{RR}\Delta RR_{j,k}$ as in Equation (12). The interquartile range (IQR) is the difference between the 75th and 25th percentiles.

Table 4 summarizes the dispersion of these components. The median relative-rank component equals -0.244 standard deviations, with an interquartile range of 0.621 standard deviations. These magnitudes indicate that rank effects are economically meaningful and vary substantially across programs and admission margins. This

result shows that declines in ordinal rank represent a quantitatively important and negative component of program value-added at selective admission margins.

5.4 Residual Academic Environment Effects

Having quantified the ordinal-rank component, we now recover the residual contribution of the academic environment. From Equation (10), the residual component for program–cutoff cell (j, k) is defined as

$$\gamma^{Q}_{j,k} = \gamma_{j,k} - \gamma^{RR} \Delta RR_{j,k}, \tag{12}$$

which subtracts the estimated rank effect from total RD value-added.

Figure 14 shows that the residual academic-environment component is shifted to the right of overall program value-added and is centered above zero. Table 4 reports a median residual effect of 0.441 standard deviations, exceeding in magnitude the median relative-rank penalty. The interquartile range of the residual component is close to one standard deviation, indicating substantial heterogeneity across programs even after accounting for rank effects.

A natural question, therefore, is what this residual component represents. Because admission to more selective programs simultaneously raises peer quality and changes institutional inputs, the residual academic-environment effect could reflect improvements in peer composition, differences in college resources, or other institutional characteristics that are bundled with selectivity. To distinguish among these channels, Table 5 relates the residual component $\gamma^{Q}_{j,k}$ to cutoff-induced changes in peer quality and fallback-relative changes in institutional characteristics. We estimate specifications that regress $\gamma^{Q}_{j,k}$ on the change in leave-own-out peer entrance scores induced by marginal admission, along with changes in program price, urban location, and institutional input indices. All right-hand-side (Δ) variables are defined as differences between the destination program and the fallback program faced by students just below the cutoff. The coefficients therefore capture how variation in cutoff-induced changes in academic environment maps into variation in residual value-added across program–cutoff cells.

Table 5 reports these results separately for competitive Computer Science Engineering (CSE) programs and for all other majors. Columns (1) and (2) relate the residual component $\gamma^{Q}_{j,k}$ to changes in peer quality and institutional characteristics alone, while Columns (3) and (4) additionally incorporate measures of institutional inputs. The regressors include the cutoff-induced change in leave-one-out peer quality, the fallback-relative change in program price (measured in units of ₹10,000), and indicators for changes in urban location. In Columns (3) and (4), we add two principal-component indices summarizing institutional inputs: a Facilities index constructed from measures

Table 5: Peer Quality and College Inputs in Residual Value-Added

	Residual Quality: $\gamma^{Q}_{j,k}$			
	CSE (1)	Non-CSE (2)	CSE (3)	Non-CSE (4)
Δ Peer Quality	-2.09**	1.49**	-1.03	1.53**
	(0.792)	(0.682)	(0.844)	(0.734)
Δ Price (10k)	0.430**	0.018	0.648***	0.033
	(0.200)	(0.166)	(0.206)	(0.186)
Δ Urban	-0.143	-0.272	-0.502	-0.238
	(0.345)	(0.372)	(0.342)	(0.430)
Δ Facilities (PCA)			-0.309**	-0.022
			(0.126)	(0.100)
Δ Teaching Resources (PCA)			0.367*	-0.049
			(0.205)	(0.139)
Constant	0.907***	0.102	0.337	0.089
	(0.299)	(0.309)	(0.357)	(0.320)
Observations	35	76	35	76
R^2	0.236	0.062	0.402	0.065

Notes: The dependent variable is the residual academic-environment value-added component, $\gamma^{Q}_{j,k}$, defined in Equation (12). Δ Peer Quality measures the change in leave-one-out average peer entrance exam scores (in standard deviations) induced by marginal admission at the cutoff. Δ Price is measured in units of ₹10,000. Δ Urban denotes fallback-relative changes in a key institutional characteristic. Δ Facilities and Δ Teaching Resources are fallback-relative changes in institutional inputs, where Facilities is the first principal component of physical infrastructure measures and Teaching Resources is the first principal component of teaching staff measures. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

of physical infrastructure (including libraries, laboratories, common rooms, computing centers, and related amenities) and a Teaching Resources index constructed from faculty counts and qualifications.[10] Each coefficient therefore captures how variation in the cutoff-induced change in a specific dimension of the academic environment is associated with variation in the residual value-added component across program–cutoff cells.

The effect of exposure to higher-achieving peers is theoretically and empirically ambiguous. One strand of literature documents positive peer spillovers in education, whereby exposure to higher-achieving classmates raises individual performance (Sacerdote, 2001; Hanushek et al., 2003). At the same time, other work emphasizes that peer

[10]The Facilities index is the first principal component of standardized measures of campus infrastructure and instructional facilities. The Teaching Resources index is the first principal component of standardized measures of faculty size and PhD-qualified teaching staff. College characteristics are obtained from publicly available AISHE (2023) datasets. The indices are defined at the destination institution and enter the regressions as fallback-relative changes induced by marginal admission.

environments may also generate competitive pressure or rank-based discouragement effects, particularly in high-ability settings (Sekhri, 2011; Murphy and Weinhardt, 2020; Fabregas, 2023). The net effect of peer composition is therefore not predetermined: stronger peers may enhance learning through complementarities, but they may also intensify competition in ways that dampen performance.

Table 5 shows that the relationship between the residual academic-environment component $\gamma^{Q}_{j,k}$ and the cutoff-induced change in peer quality differs across majors. In the highly competitive CSE programs (30% of overall enrollment and the second highest average entrance exam score), the coefficient on Δ Peer Quality is negative in Columns (1) and (3), and statistically significant in the baseline specification. Conditional on the ordinal-rank adjustment, larger improvements in peer test-score composition may not be associated with larger residual gains and may even coincide with lower net academic-environment value-added. One interpretation is that in intensely competitive cohorts, increases in peer quality amplify performance pressure or competitive dynamics that offset some of the benefits of stronger classmates. In contrast, for non-CSE programs, the peer-quality coefficient is positive and statistically significant in both specifications as evidenced by Columns (2) and (4). In these relatively less competitive majors, improvements in peer composition are associated with higher residual value-added, consistent with positive peer spillovers. The contrast between CSE and non-CSE programs suggests that the sign and magnitude of peer effects depend critically on the competitive environment of the classroom.

Program-level price differences arise within a system where tuition is formally regulated but not uniform across institutions. In many Indian states, private colleges operate within a publicly administered centralized admissions system and are affiliated to a public university. Tuition prices are set at the college or institution level, by government-appointed fee committees based on submitted cost reports and regulatory guidelines rather than market-clearing prices (Aiyer, 2022, 2025). Price differentials do not necessarily reflect differences in marginal instructional cost. They may also capture a combination of historical prestige and regulatory discretion in fee setting. As a result, price may function partly as a signal of perceived quality or excess demand rather than as a direct measure of instructional inputs.

In our decomposition framework, the cutoff-induced change in price represents a fallback-relative shift in the cost of attendance that arises from moving across institutions. Within a given college, all programs face the same regulated tuition; thus, price variation is purely at the institutional level. If regulated tuition primarily reflects institutional quality—because fee committees implicitly reward higher-quality or more prestigious colleges with higher approved prices—one would expect a positive association between Δ Price and residual quality $\gamma^{Q}_{j,k}$. Conversely, if price largely captures institutional reputation or excess demand without corresponding improvements in in-

structional inputs, its relationship with residual value-added may be weak or confined to particular high-demand contexts. Table 5 suggests the latter interpretation. While price changes are positively associated with residual value-added in CSE programs, the relationship is small and statistically insignificant in non-CSE majors. Because tuition is common across majors within an institution, this pattern implies that price proxies for institutional quality primarily in highly competitive programs, where demand and quality may be more tightly aligned. In less competitive fields, institutional price differences appear to reflect demand-side sorting or historical prestige rather than differences in academic production.

Urban location captures a salient non-academic program characteristic that may influence student demand independently of instructional quality. Programs located in urban areas are often preferred for reasons related to amenities, labor market access, or proximity to home rather than academic production. The coefficients in Table 5 are negative and imprecise, suggesting that urban location does not systematically translate into higher residual academic-environment value-added. The estimates are consistent with urban status reflecting demand-side preferences rather than improvements in instructional inputs.

We next turn to direct measures of institutional inputs. Unlike price and urban status, the Facilities and Teaching Resources indices are constructed to capture observable production inputs. The Facilities index summarizes physical infrastructure such as laboratories, computing centers, and campus amenities, while the Teaching Resources index captures faculty size and qualifications through a principal component of staffing measures. If residual value-added reflects improvements in academic production, one would expect positive associations with these input measures. Table 5 yields an intriguing pattern for physical infrastructure. In CSE programs, the Facilities index is negatively and statistically significant, whereas in non-CSE majors the association is small and imprecisely estimated. While the limited sample size warrants caution in interpretation, this pattern may suggest that greater physical infrastructure does not automatically translate into higher marginal academic gains. Infrastructure may proxy for institutional scale or prestige rather than instructional effectiveness, in which case capital investments alone may not drive residual value-added at the margin. The Teaching Resources index yields somewhat different patterns. In CSE programs, the coefficient on Δ Teaching Resources is positive and statistically significant for CSE while for non-CSE majors the estimates are smaller and imprecise. This pattern is consistent with the possibility that instructional inputs—captured by faculty size and qualifications—may matter more in highly competitive academic environments. In settings where peer composition and competitive pressure are pronounced, marginal improvements in teaching resources may help sustain academic performance and offset some of the negative effects associated with rank declines. By contrast, in less

competitive majors, variation in teaching inputs appears less tightly linked to marginal value-added. These results tentatively suggest that instructional quality, rather than physical infrastructure, may play a more meaningful role in shaping residual academic gains in competitive programs.

These results indicate that the residual academic-environment component cannot be attributed solely to changes in peer test-score composition. Nor does it appear to reflect purely demand-side characteristics such as urban location or regulated price differences. Instead, the evidence suggests that the residual captures a combination of instructional inputs and competitive peer dynamics that operate differently across programs. In highly competitive fields such as CSE, peer composition and teaching resources appear closely intertwined with competitive pressures, while physical infrastructure plays a more limited role. In less competitive majors, peer spillovers operate more conventionally and instructional inputs are less tightly linked to marginal gains. Overall, the residual value-added component reflects meaningful institutional mechanisms beyond ordinal rank effects, highlighting that the academic returns to selectivity depend not only on relative standing but also on the competitive and instructional environment of the program.

6 Conclusion

This paper estimates the academic value-added of selective college–major programs in a large Indian higher-education system. Leveraging centralized admissions with program-by-category cutoffs and standardized exit examinations, we implement a regression discontinuity (RD) design that identifies the local effect of marginal admission into more selective programs. We find that crossing an admission cutoff increases standardized academic performance by roughly 0.11 standard deviations and raises graduation rates by 3–4 percentage points. These effects are stable across bandwidths and disciplines, indicating positive academic gains at the admission margin.

We show that this RD effect reflects the net impact of two opposing forces that are jointly determined by program assignment. Admission to a more selective program substantially improves peer quality and alters exposure to institutional resources, while simultaneously lowering a student's ordinal rank within the admitted cohort. Exploiting multiple category-specific cutoffs within the same program, we isolate the ordinal-rank channel and find that declines in relative standing reduce academic performance in economically meaningful ways. However, the gains associated with improvements in academic environment dominate the offsetting penalty from lower relative rank on average. Program value-added therefore emerges as the net result of distinct mechanisms, potentially operating in opposite directions.

Our results have implications for both measurement and policy. Observational

value-added models, commonly used in research and institutional rankings, can summarize different counterfactual comparisons when students enter programs through heterogeneous admission margins. In particular, when disadvantaged students enter at substantially lower ordinal ranks, observational estimates may partially reflect rank penalties in addition to institutional quality. This sensitivity has direct implications for interpreting evidence in affirmative-action settings. While observational patterns may suggest negative effects, margin-specific estimates reveal positive academic gains at the cutoff.

More broadly, our findings underscore that the margin at which students enter programs fundamentally shapes what value-added measures capture. Access to more selective programs generates positive academic returns at the margin, even when accompanied by declines in relative standing within the cohort. By decomposing program value-added into academic-environment and ordinal-rank components, we provide a framework for interpreting heterogeneous estimates of institutional quality in systems with centralized admissions and reservation policies. This perspective clarifies how program quality should be measured when admission margins differ across students.

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Appendix

A Derivation of the Rank–Quality Decomposition

This appendix provides a formal algebraic derivation of the decomposition of regression-discontinuity (RD) program value-added into ordinal rank and residual quality components. We show explicitly how multiple admission cutoffs within the same program allow the ordinal rank effect to be separately identified under transparent assumptions.

A.1 Notation and Setup

Let i index students, j index college–major programs, and k index seat categories within a program (e.g., Open Category (OC) and Affirmative Action (AA)). Each admission margin is defined by a program–category pair $c = (j, k)$.

Let:

- Y_{ij} denote a standardized academic outcome for student i enrolled in program j.
- RR_{ij} denote student i's ordinal (percentile) rank within program j.
- Q_j denote a composite index of the academic environment of program j, encompassing institutional inputs and peer composition.
- A_i denote student ability and other time-invariant determinants of achievement.

As in Section 3.4, let $\gamma_{j,k}$ denote the fuzzy RD-IV estimand associated with crossing cutoff k for program j, interpreted as the causal effect of enrolling in program j for students at that margin.

A.2 Structural Outcomes and First Differencing

We begin from the structural outcome equation:

$$Y_{ij} = \gamma^Q Q_j + \gamma^{RR} RR_{ij} + A_i + \varepsilon_{ij}. \tag{13}$$

Consider a student at admission margin (j, k). Let $\neg(j, k)$ denote the fallback program assigned to students who do not enroll in j at cutoff k. Outcomes under admission and fallback are:

$$Y_{ij} = \gamma^Q Q_j + \gamma^{RR} RR_{ij} + A_i + \varepsilon_{ij}, \tag{14}$$

$$Y_{i\neg(j,k)} = \gamma^Q Q_{\neg(j,k)} + \gamma^{RR} RR_{i\neg(j,k)} + A_i + \varepsilon_{i\neg(j,k)}. \tag{15}$$

Taking the difference yields:

$$Y_{ij} - Y_{i\neg(j,k)} = \gamma^{Q}\left(Q_j - Q_{\neg(j,k)}\right) + \gamma^{RR}\left(RR_{ij} - RR_{i\neg(j,k)}\right) + \left(\varepsilon_{ij} - \varepsilon_{i\neg(j,k)}\right). \tag{16}$$

Under the standard RD smoothness assumption, A_i and the distribution of ε_{ij} vary smoothly at the cutoff, so their expected difference vanishes for compliers. Taking expectations conditional on compliance yields the RD estimand:

$$\gamma_{j,k} = \gamma^{Q}\Delta Q_{j,k} + \gamma^{RR}\Delta RR_{j,k}, \tag{17}$$

where

$$\Delta Q_{j,k} := Q_j - Q_{\neg(j,k)}, \qquad \Delta RR_{j,k} := RR_{ij} - RR_{i\neg(j,k)}.$$

A.3 Second Differencing Across Admission Margins

Fix a program j with two admission margins $k \in \{\text{OC}, \text{AA}\}$. The RD estimands at these margins satisfy:

$$\gamma_{j,\text{OC}} = \gamma^{Q}\Delta Q_{j,\text{OC}} + \gamma^{RR}\Delta RR_{j,\text{OC}}, \tag{18}$$

$$\gamma_{j,\text{AA}} = \gamma^{Q}\Delta Q_{j,\text{AA}} + \gamma^{RR}\Delta RR_{j,\text{AA}}. \tag{19}$$

Taking the difference across margins:

$$\gamma_{j,\text{OC}} - \gamma_{j,\text{AA}} = \gamma^{Q}\left(\Delta Q_{j,\text{OC}} - \Delta Q_{j,\text{AA}}\right) + \gamma^{RR}\left(\Delta RR_{j,\text{OC}} - \Delta RR_{j,\text{AA}}\right). \tag{20}$$

Assumption A1 (Within-program fallback invariance). For a given destination program j,

$$\mathbb{E}\left[\Delta Q_{j,\text{OC}} - \Delta Q_{j,\text{AA}}\right] = 0.$$

That is, students who miss OC and AA cutoffs face fallback programs with similar academic environments in expectation.

Under Assumption A1, the quality term cancels in (20), yielding:

$$\gamma_{j,\text{OC}} - \gamma_{j,\text{AA}} = \gamma^{RR}\left(\Delta RR_{j,\text{OC}} - \Delta RR_{j,\text{AA}}\right). \tag{21}$$

Equation (21) shows that within-program differences in RD value-added across admission margins identify the ordinal rank effect.

A.4 Estimation and Component Construction

Aggregating across programs, γ^{RR} may be estimated using either a Wald-style estimator:

$$\widehat{\gamma^{RR}} = \frac{\sum_j w_j \left(\widehat{\gamma}_{j,\mathrm{OC}} - \widehat{\gamma}_{j,\mathrm{AA}}\right)}{\sum_j w_j \left(\widehat{\Delta RR}_{j,\mathrm{OC}} - \widehat{\Delta RR}_{j,\mathrm{AA}}\right)}, \tag{22}$$

or equivalently via the regression:

$$\widehat{\gamma}_{j,k} = \mu_j + \gamma^{RR}\widehat{\Delta RR}_{j,k} + e_{j,k}, \tag{23}$$

where μ_j are program fixed effects.

Given $\widehat{\gamma^{RR}}$, the rank and residual quality components are defined as:

$$\widehat{\gamma}^{RR}_{j,k} := \widehat{\gamma^{RR}} \cdot \widehat{\Delta RR}_{j,k}, \tag{24}$$

$$\widehat{\gamma}^{Q}_{j,k} := \widehat{\gamma}_{j,k} - \widehat{\gamma}^{RR}_{j,k}. \tag{25}$$

A.5 Interpretation and Scope

The residual component $\widehat{\gamma}^{Q}_{j,k}$ captures the combined contribution of institutional inputs and peer quality net of ordinal rank effects. The framework deliberately refrains from further decomposing this residual. Its purpose is to isolate the rank channel transparently and to clarify how competing mechanisms jointly shape program value-added estimates.

B Additional Tables and Figures

Table B1: First Stage – Probability of Enrollment (±10 Bandwidth)

	$\mathbb{P}$(Enrollment = 1)
$\mathbb{1}\{X_{ijk} \geq 0\}$	0.879***
	(0.009)
N	17,297
R^2	0.717

Notes: This table reports the first-stage RD estimate of the effect of admission eligibility on enrollment using a ±10 exam-score bandwidth. The coefficient is the discontinuity at the cutoff from a local linear RD specification. Standard errors (in parentheses) are clustered at the program level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B2: Covariate Balance at Admission Cutoffs (±10 Bandwidth)

	12th Marks (marks)	Student Age (years)	Out-of-pocket Expense (Rupees)
$\mathbb{1}\{X_{ijk} \geq 0\}$	0.074	-0.016	1,231.4
	(0.201)	(0.016)	(1,267.7)
N	17,297	17,297	17,297

Notes: This table reports RD balance tests for predetermined covariates at program admission cutoffs using a ±10 entrance examination score bandwidth. Each column reports the estimated discontinuity at the cutoff from a local linear RD specification. Standard errors (in parentheses) are clustered at the program level. None of the covariates exhibits a statistically significant discontinuity, supporting the validity of the RD design.

Table B3: Peer Quality and Relative Percentile Rank: ±10 Bandwidth

	Peer Quality		Relative Rank	
	All (SD) (1)	Affiliated (SD) (2)	All (p.p.) (3)	Affiliated (p.p.) (4)
$\mathbb{1}\{X_{ijk} \geq 0\}$	0.376***	0.366***	-18.1***	-17.3***
	(0.022)	(0.022)	(1.08)	(1.11)
N	17,297	14,543	17,297	14,543
R^2	0.90064	0.87874	0.12995	0.14048

Note: RD estimates of the effect of marginal admission on peer quality (average peer entrance exam score) and relative percentile rank. Enrolled students, on average, experience statistically significant increases in peer quality and decreases in relative percentile rank. Standard errors (in parentheses) are clustered at the program level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B4: Program Value-Added Heterogeneity in CSE Branch

	25th pct.	50th pct.	75th pct.	IQR
Panel A: OLS Value-Added				
Overall	-0.157	0.039	0.235	0.392
GEN	-0.098	0.004	0.199	0.298
BC	-0.720	-0.487	-0.272	0.448
SC/ST	-0.964	-0.479	0.082	1.046
Panel B: RD Value-Added				
Overall	-0.106	0.136	0.506	0.611
GEN	-0.140	0.072	0.265	0.405
BC	-0.206	0.092	0.397	0.603
SC/ST	-0.249	-0.085	0.295	0.544

Note: Table reports the 25th, 50th, and 75th percentiles of program value-added (VA) distributions in the CSE branch, along with the interquartile range (IQR = $75^{th} - 25^{th}$). Results are shown separately for OLS and RD estimates, overall and by caste group.

Table B5: Regression Discontinuity Estimates of Academic Outcomes by Bandwidth (CSE Only)

	±6 (1)	±10 (2)	±14 (3)	±18 (4)
Total Marks (SD)	0.116**	0.110**	0.106**	0.112***
	(0.057)	(0.045)	(0.043)	(0.039)
$\mathbb{P}$(Graduation = 1)	0.037**	0.035**	0.030**	0.031**
	(0.018)	(0.015)	(0.013)	(0.013)
Year 2 Marks (SD)	0.098*	0.088*	0.083*	0.092**
	(0.056)	(0.044)	(0.043)	(0.039)
Year 3 Marks (SD)	0.122**	0.124***	0.118***	0.121***
	(0.055)	(0.043)	(0.043)	(0.040)
Year 4 Marks (SD)	0.124**	0.112**	0.097**	0.103**
	(0.056)	(0.045)	(0.041)	(0.039)
N	4,319	5,482	5,919	6,109

Notes: Each cell reports a two-stage least squares (2SLS) estimate from a fuzzy regression discontinuity design, where enrollment in the allotted program is instrumented by admission eligibility (crossing the program cutoff). Estimates are shown for Computer Science and Engineering (CSE) programs only, using alternative entrance-exam score bandwidths around the cutoff. Outcomes are standardized within cohort. Standard errors (in parentheses) are clustered at the program level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B6: Value-Added and Ordinal Rank Effects by Program-Cutoff Type

	25th pct.	50th pct.	75th pct.	IQR
$\gamma_{j,OC}$	-0.217	-0.051	0.186	0.403
$\gamma_{j,AA}\gamma_{j,AA}$	-0.164	0.102	0.499	0.663
$\gamma_{j,OC} - \gamma_{j,AA}$	-0.457	-0.131	0.143	0.600
$\Delta RR_{j,OC}$	-0.752	-0.636	-0.442	0.311
$\Delta RR_{j,AA}$	-0.610	-0.277	-0.141	0.469
$\Delta RR_{j,OC} - \Delta RR_{j,AA}$	-0.472	-0.315	-0.023	0.449

Notes: This table reports selected percentiles of the distributions of program value-added and decline in relative rank decline distributions disaggregated by program-cutoff type. $\gamma_{j,k}$ denotes the RD value-added at admission cutoff k for program j, where $k \in \{OC, AA\}$. $\Delta RR_{j,k}$ terms denote the change in ordinal rank experienced by a marginally admitted student who clears cutoff-type k and is admitted to a program. The differences in program value-added and relative rank effects by cutoff type are defined in Equation (11). The interquartile range (IQR) is the difference between the 75th and 25th percentiles.

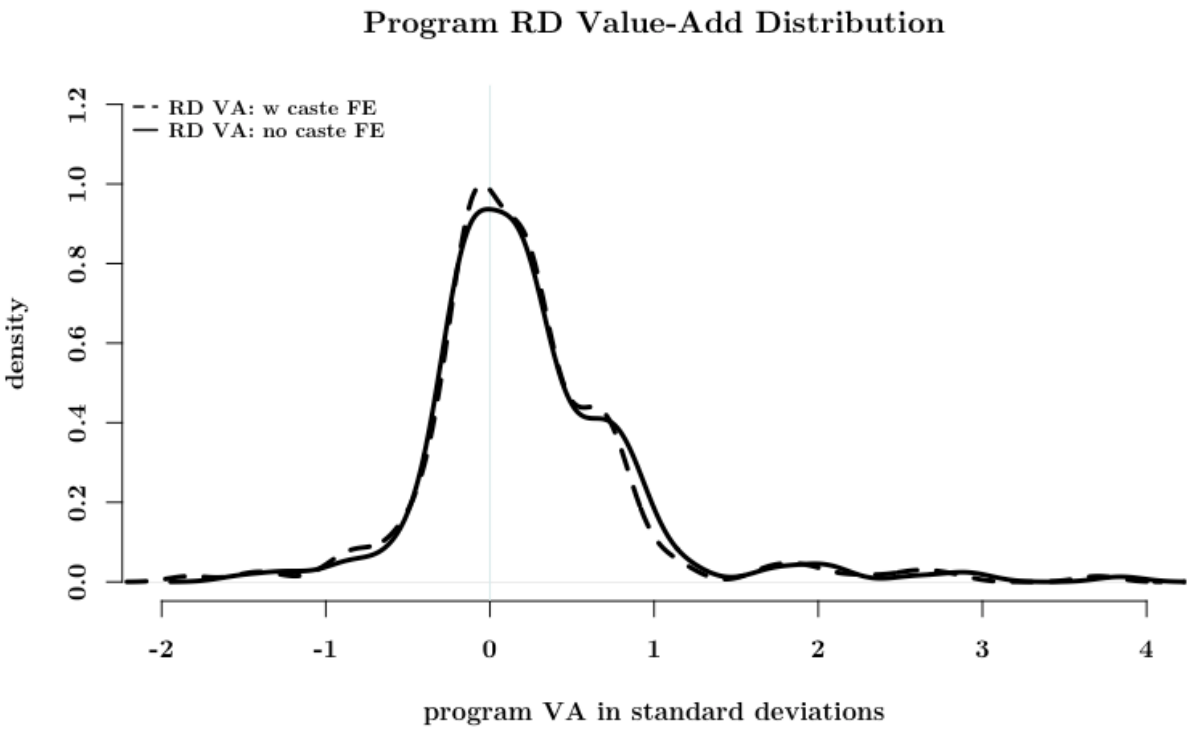

Figure B1: Program RD Value-Added Estimates With and Without Caste Fixed Effects

Notes: This figure plots the distribution of program-level RD value-added estimates (standard deviations). The dashed line reports estimates from specifications without caste fixed effects, while the solid line reports estimates that include caste fixed effects. The similarity of the distributions indicates that RD value-added estimates are not sensitive to the inclusion of caste controls.